MP10 centers, neighborhoods and formal fronts

63 sections | 265,002 tissue spots | 143,461 frozen Epi queries

57 primary programs + 23 sensitivity programs; original Top20 AUCell unchanged.

5,664 P95 tumor-Epi reference points; 1,979 exact-front-position matched pairs.

R0: modest stress/growth associations; shared-gene removal attenuates several signals.

Androgen and TGF-beta associations largely disappear without MP10 overlap genes.

p-EMT is modestly lower at MP10-high centers; spatial effects weaken with distance.

No consistent global p-EMT / Flores increase at the formal v3.4 front.

No robust P95 front enrichment or MP10-specific R12 preference after family FDR.

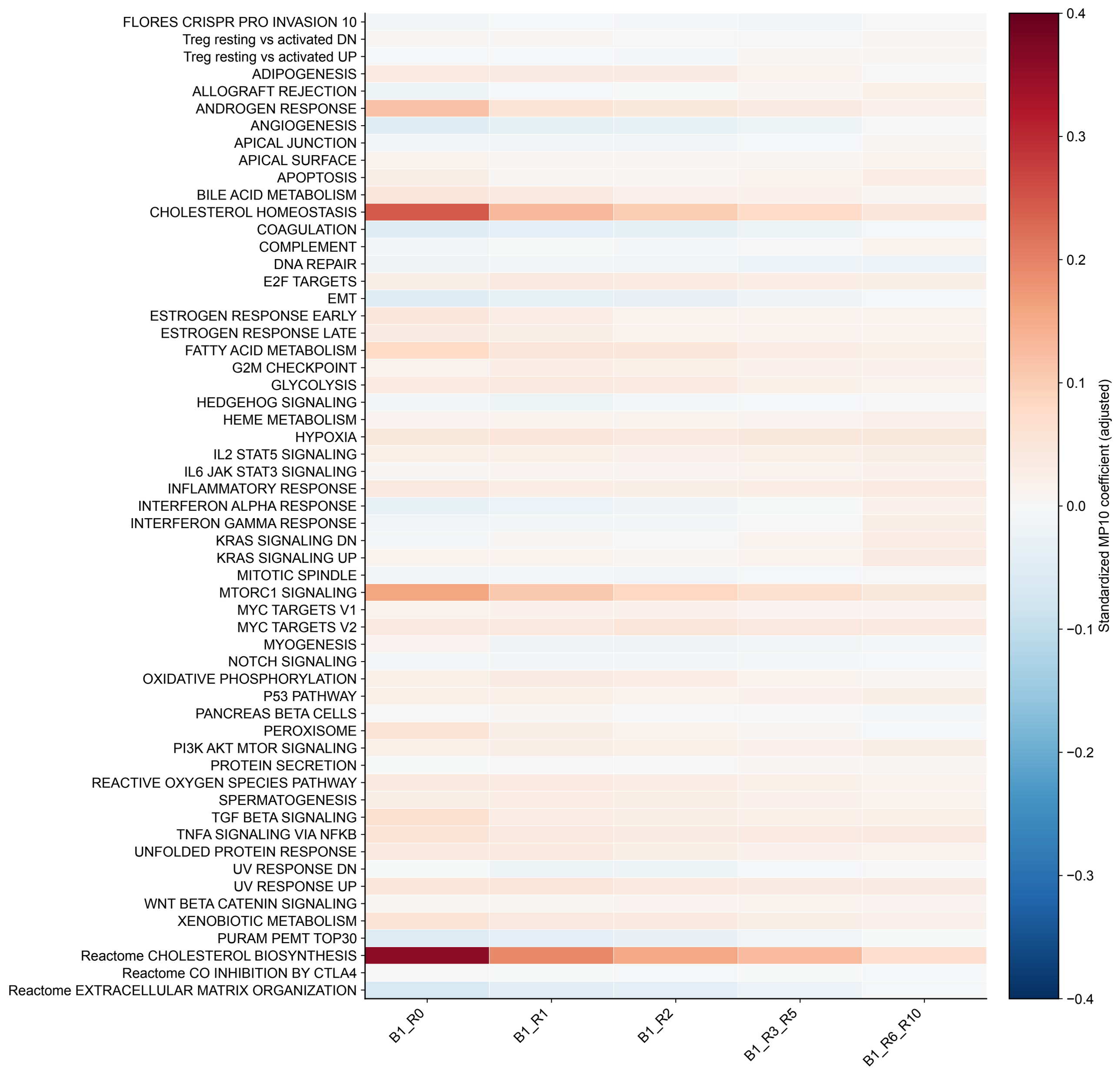
Main repeated-unit summary: 54 units / 6 data sets; verified subset: 27 patients / 3 sets.

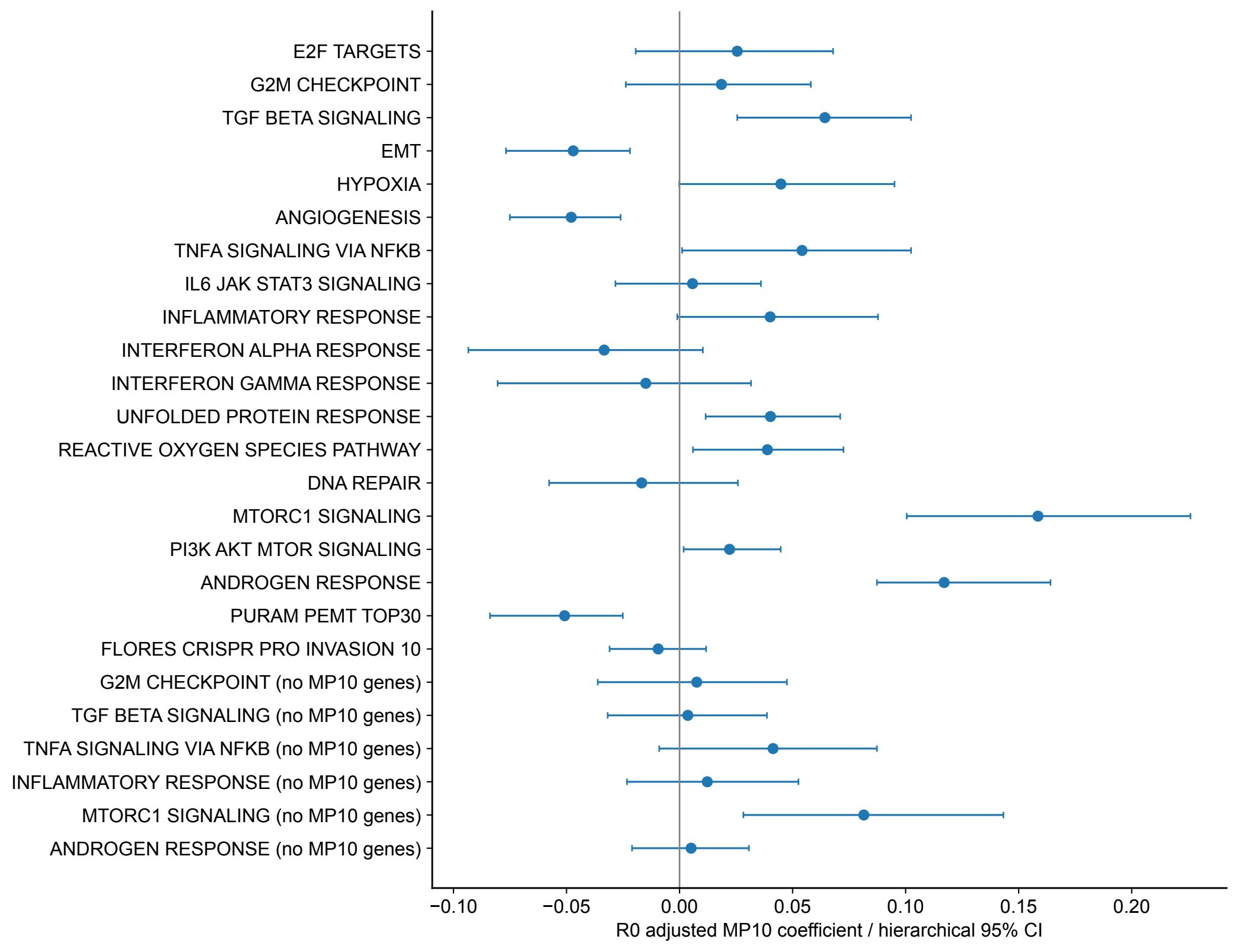
Normal-source samples and unverified identities remain separate.

All effects, uncertainty, coverage, negative findings and missing modules are retained.

Read summary.md and methods.md before biological interpretation.
PDF graphics are vector; H&E retains its original raster format.


CanDis__UNVERIFIED
GSE171351
GSE238145
GSE285715
GSE319536
Inhousedata__UNVERIFIED
NPJ__UNVERIFIED
13 13 13 13 11 13
4 3 4 3 2 3
10 9 9 9 7 9
6 6 3 3 3 6
16 16 16 15 14 16
7 7 7 7 6 7
7 7 7 7 5 7
registered
Epi
P95
strict_front
front_match
R12
Number of sections
16
14
12
10
8
6
4
2


FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
APOPTOSIS
BILE ACID METABOLISM
CHOLESTEROL HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA REPAIR
E2F TARGETS
EMT
ESTROGEN RESPONSE EARLY
ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
G2M CHECKPOINT
GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
IL2 STAT5 SIGNALING
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
MTORC1 SIGNALING
MYC TARGETS V1
MYC TARGETS V2
MYOGENESIS
NOTCH SIGNALING
OXIDATIVE PHOSPHORYLATION
P53 PATHWAY
PANCREAS BETA CELLS
PEROXISOME
PI3K AKT MTOR SIGNALING
PROTEIN SECRETION
REACTIVE OXYGEN SPECIES PATHWAY
SPERMATOGENESIS
TGF BETA SIGNALING
TNFA SIGNALING VIA NFKB
UNFOLDED PROTEIN RESPONSE
UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
PURAM PEMT TOP30
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION
B1_R0
B1_R1
B1_R2
B1_R3_R5
B1_R6_R10
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Standardized MP10 coefficient (adjusted)


E2F TARGETS
G2M CHECKPOINT
TGF BETA SIGNALING
EMT
HYPOXIA
ANGIOGENESIS
TNFA SIGNALING VIA NFKB
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
UNFOLDED PROTEIN RESPONSE
REACTIVE OXYGEN SPECIES PATHWAY
DNA REPAIR
MTORC1 SIGNALING
PI3K AKT MTOR SIGNALING
ANDROGEN RESPONSE
PURAM PEMT TOP30
FLORES CRISPR PRO INVASION 10
G2M CHECKPOINT (no MP10 genes)
TGF BETA SIGNALING (no MP10 genes)
TNFA SIGNALING VIA NFKB (no MP10 genes)
INFLAMMATORY RESPONSE (no MP10 genes)
MTORC1 SIGNALING (no MP10 genes)
ANDROGEN RESPONSE (no MP10 genes)
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
R0 adjusted MP10 coefficient / hierarchical 95% CI

FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
APOPTOSIS
BILE ACID METABOLISM
CHOLESTEROL HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA REPAIR
E2F TARGETS
EMT
ESTROGEN RESPONSE EARLY
ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
G2M CHECKPOINT
GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
IL2 STAT5 SIGNALING
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
MTORC1 SIGNALING
MYC TARGETS V1
MYC TARGETS V2
MYOGENESIS
NOTCH SIGNALING
OXIDATIVE PHOSPHORYLATION
P53 PATHWAY
PANCREAS BETA CELLS
PEROXISOME
PI3K AKT MTOR SIGNALING
PROTEIN SECRETION
REACTIVE OXYGEN SPECIES PATHWAY
SPERMATOGENESIS
TGF BETA SIGNALING
TNFA SIGNALING VIA NFKB
UNFOLDED PROTEIN RESPONSE
UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
PURAM PEMT TOP30
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION

B4_matched_front_R0
B4_matched_front_R1
B4_matched_front_R2
B4_matched_front_R3_R5
B4_matched_front_R6_R10

0.4
0.2
0.0
−0.2
−0.4
P95 minus matched control (section SD)

FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
APOPTOSIS
BILE ACID METABOLISM
CHOLESTEROL HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA REPAIR
E2F TARGETS
EMT
ESTROGEN RESPONSE EARLY
ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
G2M CHECKPOINT
GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
IL2 STAT5 SIGNALING
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
MTORC1 SIGNALING
MYC TARGETS V1
MYC TARGETS V2
MYOGENESIS
NOTCH SIGNALING
OXIDATIVE PHOSPHORYLATION
P53 PATHWAY
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PI3K AKT MTOR SIGNALING
PROTEIN SECRETION
REACTIVE OXYGEN SPECIES PATHWAY
SPERMATOGENESIS
TGF BETA SIGNALING
TNFA SIGNALING VIA NFKB
UNFOLDED PROTEIN RESPONSE
UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
PURAM PEMT TOP30
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION

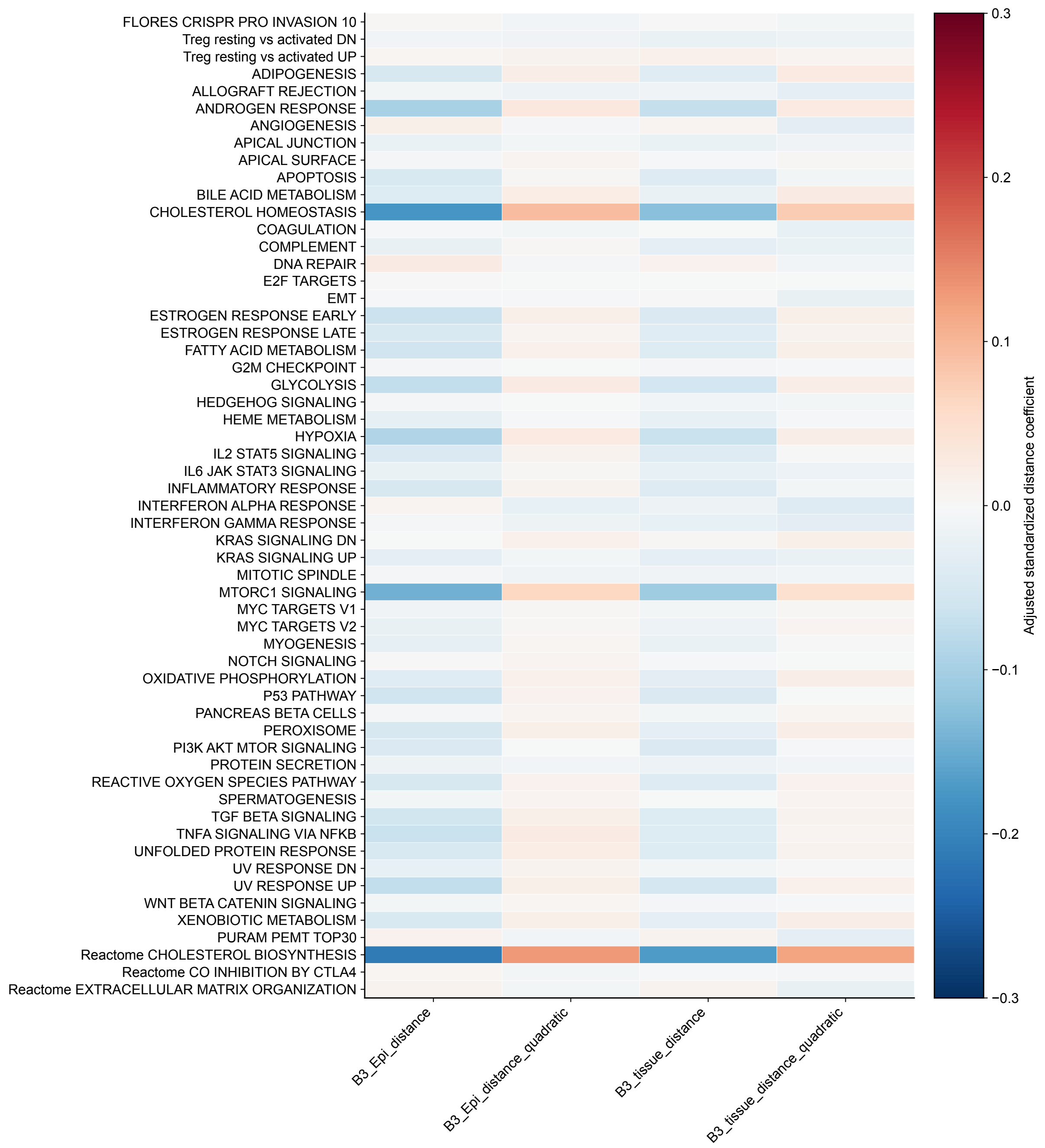
R0
R1
R2
R3_R5
R6_R10

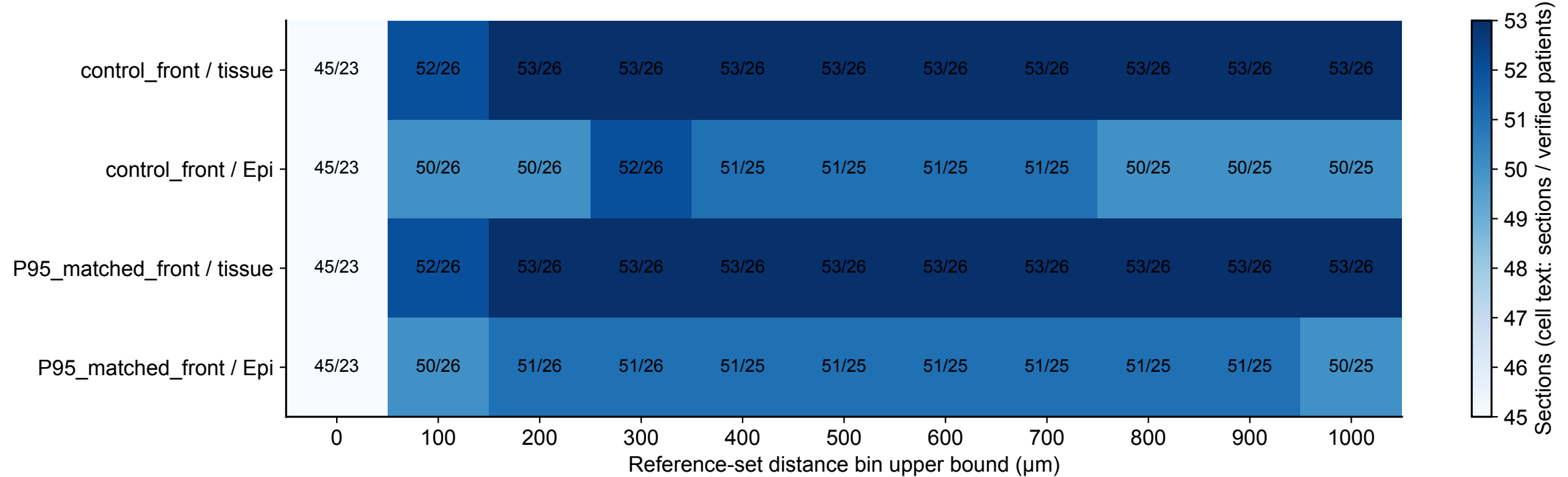
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Row z-score of mean UCell / Epi

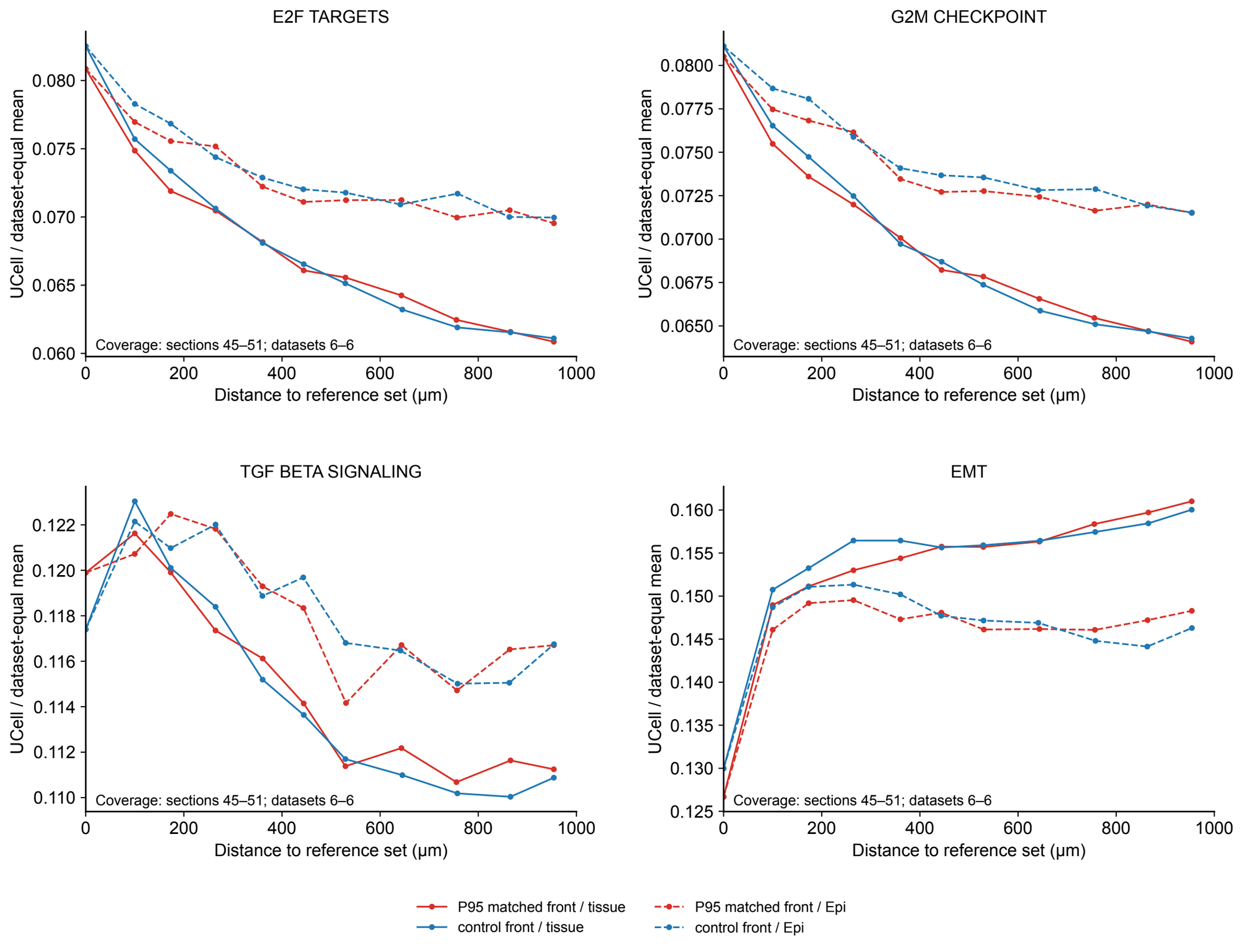
FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
APOPTOSIS
BILE ACID METABOLISM
CHOLESTEROL HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA REPAIR
E2F TARGETS
EMT
ESTROGEN RESPONSE EARLY
ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
G2M CHECKPOINT
GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
IL2 STAT5 SIGNALING
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
MTORC1 SIGNALING
MYC TARGETS V1
MYC TARGETS V2
MYOGENESIS
NOTCH SIGNALING
OXIDATIVE PHOSPHORYLATION
P53 PATHWAY
PANCREAS BETA CELLS
PEROXISOME
PI3K AKT MTOR SIGNALING
PROTEIN SECRETION
REACTIVE OXYGEN SPECIES PATHWAY
SPERMATOGENESIS
TGF BETA SIGNALING
TNFA SIGNALING VIA NFKB
UNFOLDED PROTEIN RESPONSE
UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
PURAM PEMT TOP30
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION

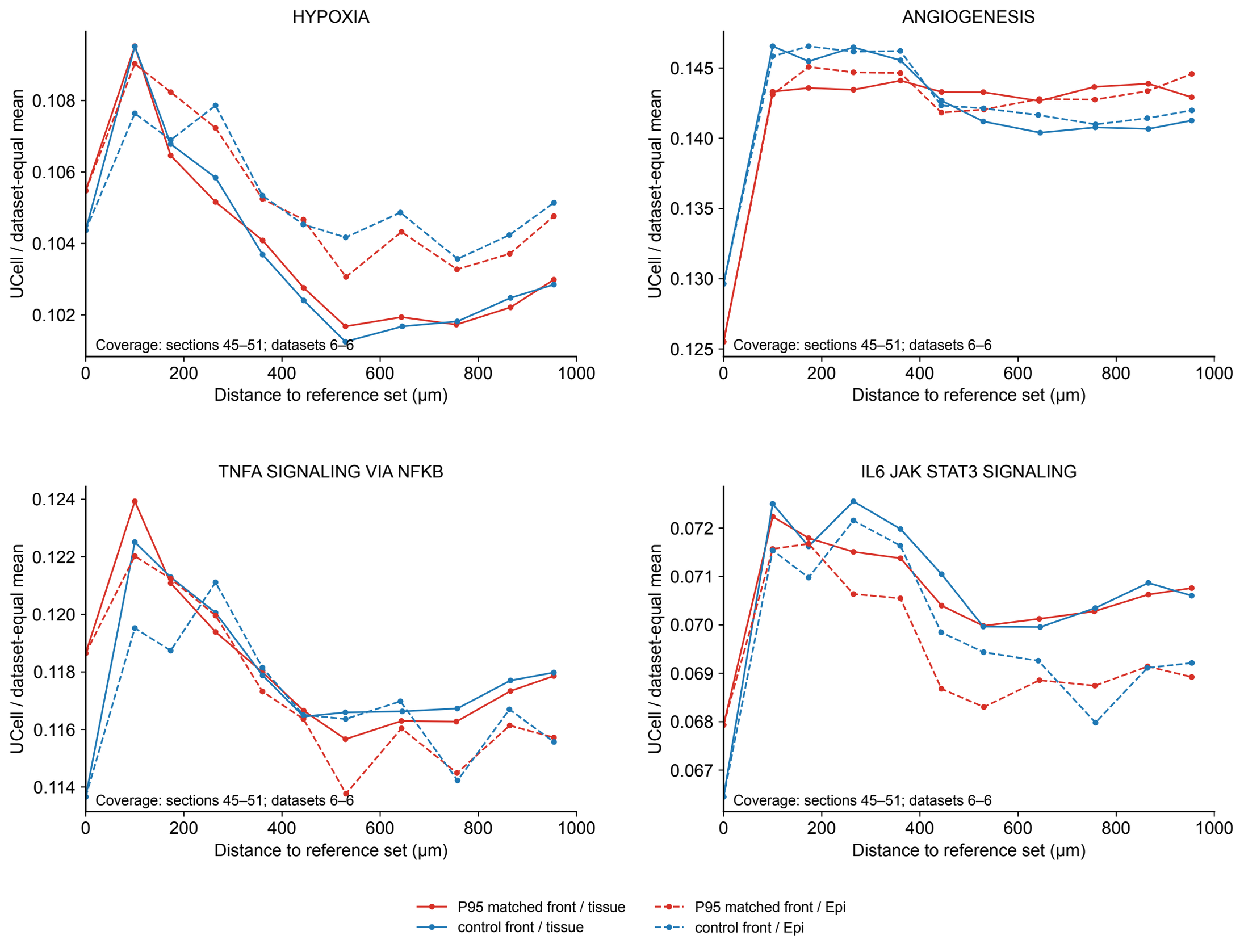
R0
R1
R2
R3_R5
R6_R10

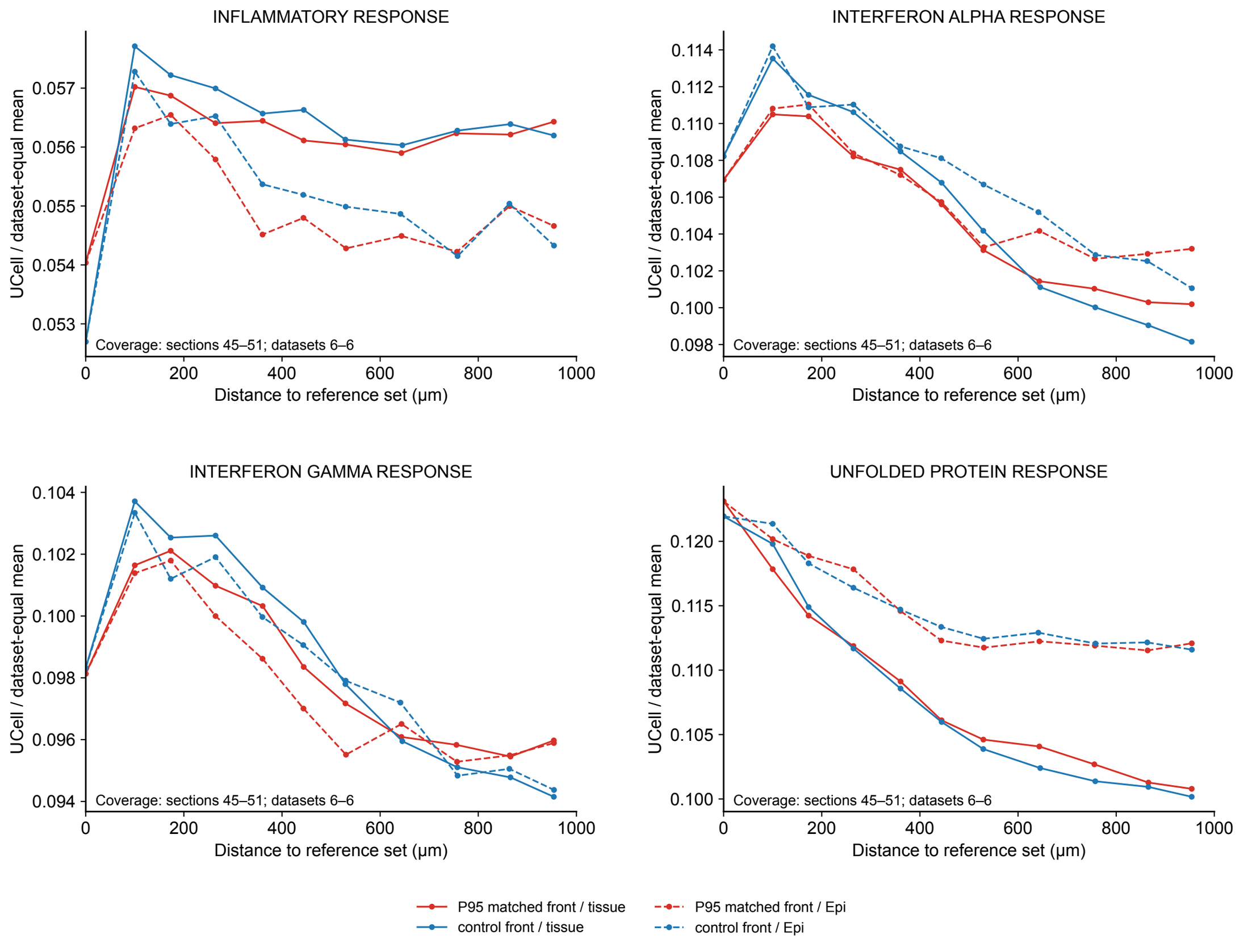
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Row z-score of mean UCell / tissue

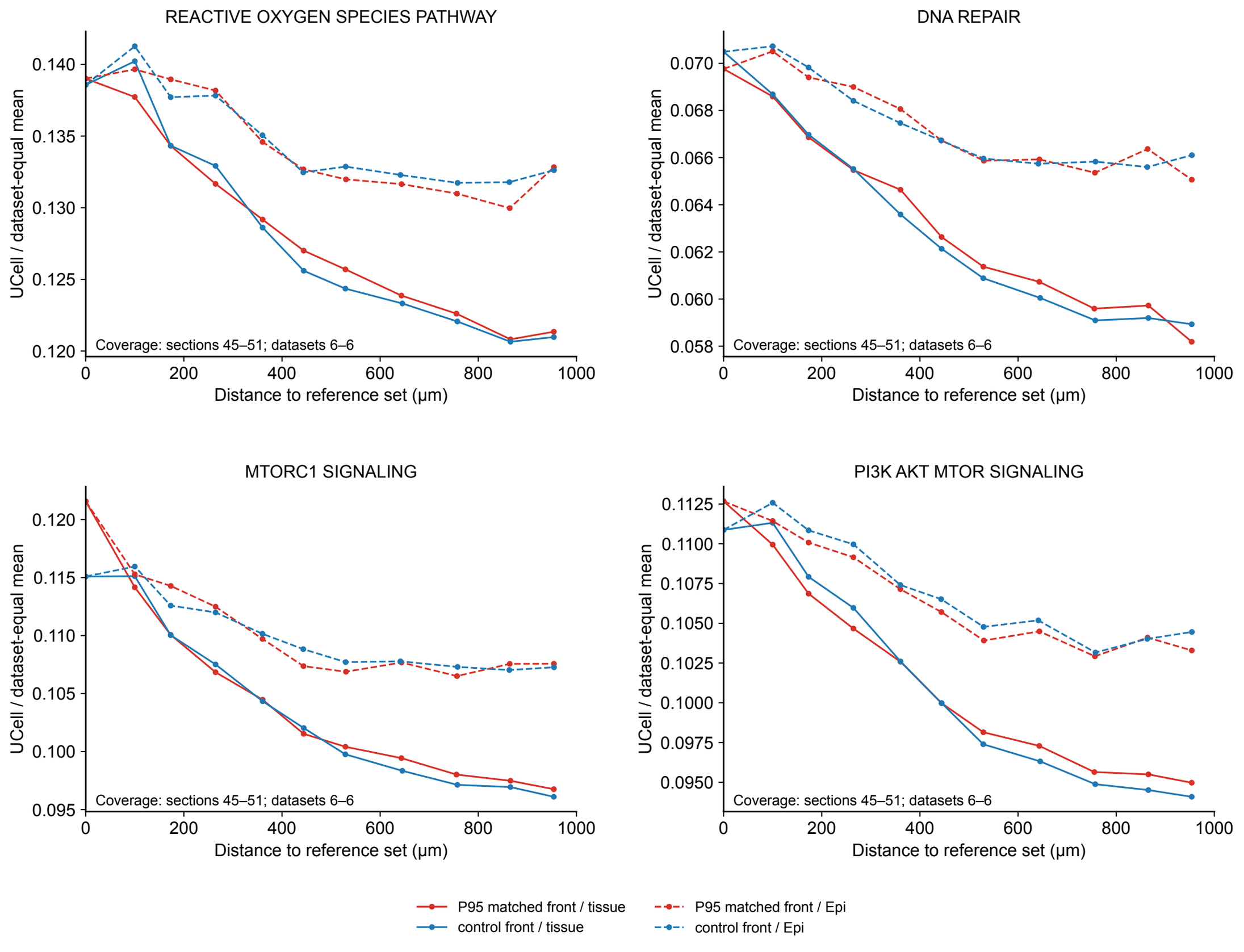

FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
APOPTOSIS
BILE ACID METABOLISM
CHOLESTEROL HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA REPAIR
E2F TARGETS
EMT
ESTROGEN RESPONSE EARLY
ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
G2M CHECKPOINT
GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
IL2 STAT5 SIGNALING
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
MTORC1 SIGNALING
MYC TARGETS V1
MYC TARGETS V2
MYOGENESIS
NOTCH SIGNALING
OXIDATIVE PHOSPHORYLATION
P53 PATHWAY
PANCREAS BETA CELLS
PEROXISOME
PI3K AKT MTOR SIGNALING
PROTEIN SECRETION
REACTIVE OXYGEN SPECIES PATHWAY
SPERMATOGENESIS
TGF BETA SIGNALING
TNFA SIGNALING VIA NFKB
UNFOLDED PROTEIN RESPONSE
UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
PURAM PEMT TOP30
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION
B3_Epi_distance
B3_Epi_distance_quadratic
B3_tissue_distance
B3_tissue_distance_quadratic
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Adjusted standardized distance coefficient

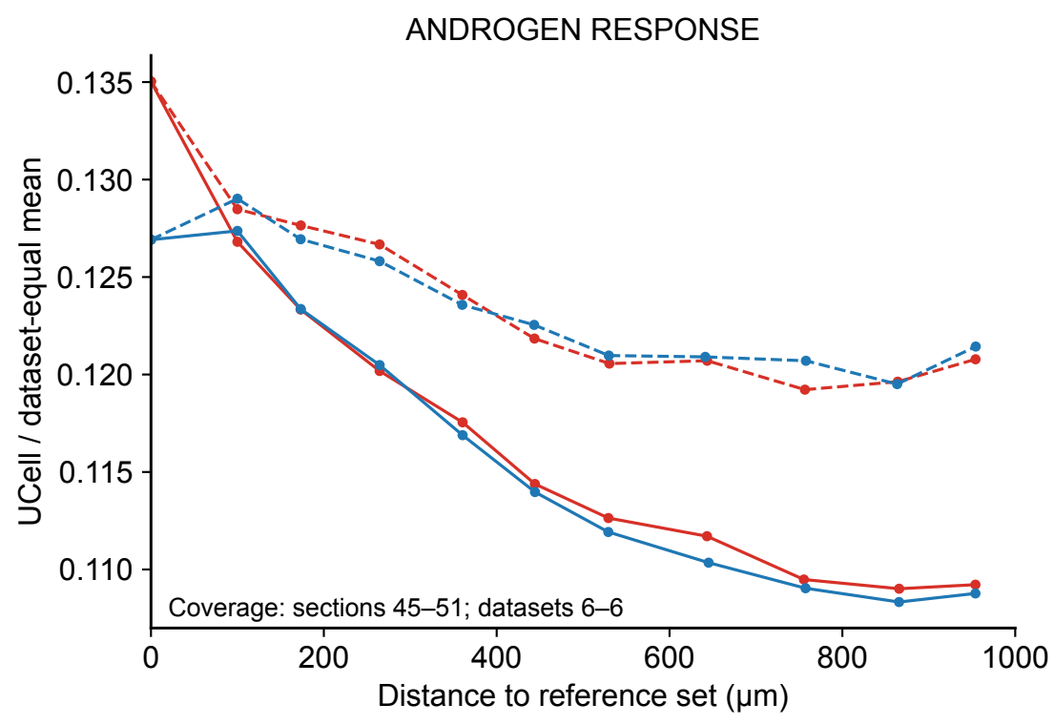

control_front / tissue
control_front / Epi
P95_matched_front / tissue
P95_matched_front / Epi
45/23
52/26
53/26
53/26
53/26
53/26
53/26
53/26
53/26
53/26
53/26
45/23
50/26
50/26
52/26
51/25
51/25
51/25
51/25
50/25
50/25
50/25
45/23
52/26
53/26
53/26
53/26
53/26
53/26
53/26
53/26
53/26
53/26
45/23
50/26
51/26
51/26
51/25
51/25
51/25
51/25
51/25
51/25
50/25
0
100
200
300
400
500
600
700
800
900
1000
Reference-set distance bin upper bound (µm)
53
52
51
50
49
48
47
46
45
Sections (cell text: sections / verified patients)

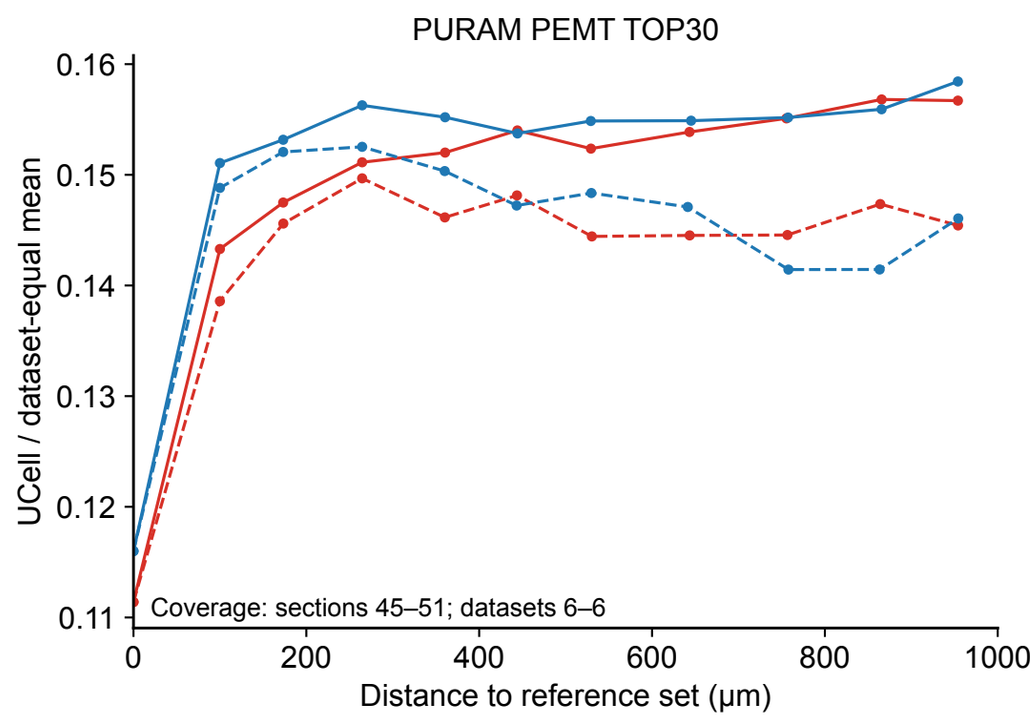

E2F TARGETS
G2M CHECKPOINT
TGF BETA SIGNALING
EMT
UCell / dataset-equal mean
Distance to reference set (μm)
Coverage: sections 45–51; datasets 6–6
P95 matched front / tissue
control front / tissue
P95 matched front / Epi
control front / Epi

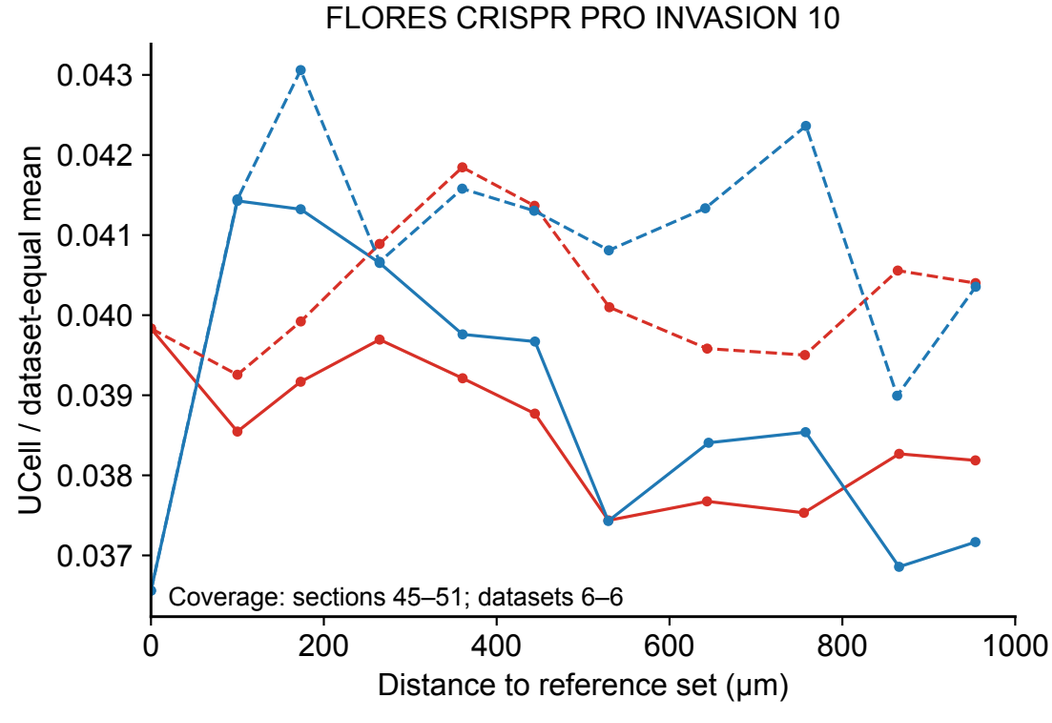

HYPOXIA
ANGIOGENESIS
TNFA SIGNALING VIA NFKB
IL6 JAK STAT3 SIGNALING
UCell / dataset-equal mean
Distance to reference set (μm)
Coverage: sections 45–51; datasets 6–6
P95 matched front / tissue
control front / tissue
P95 matched front / Epi
control front / Epi

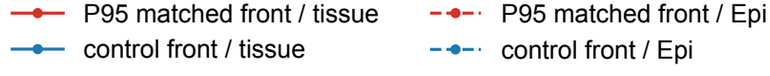

INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
UNFOLDED PROTEIN RESPONSE
UCell / dataset-equal mean
Distance to reference set (µm)
Coverage: sections 45–51; datasets 6–6
P95 matched front / tissue
control front / tissue
P95 matched front / Epi
control front / Epi

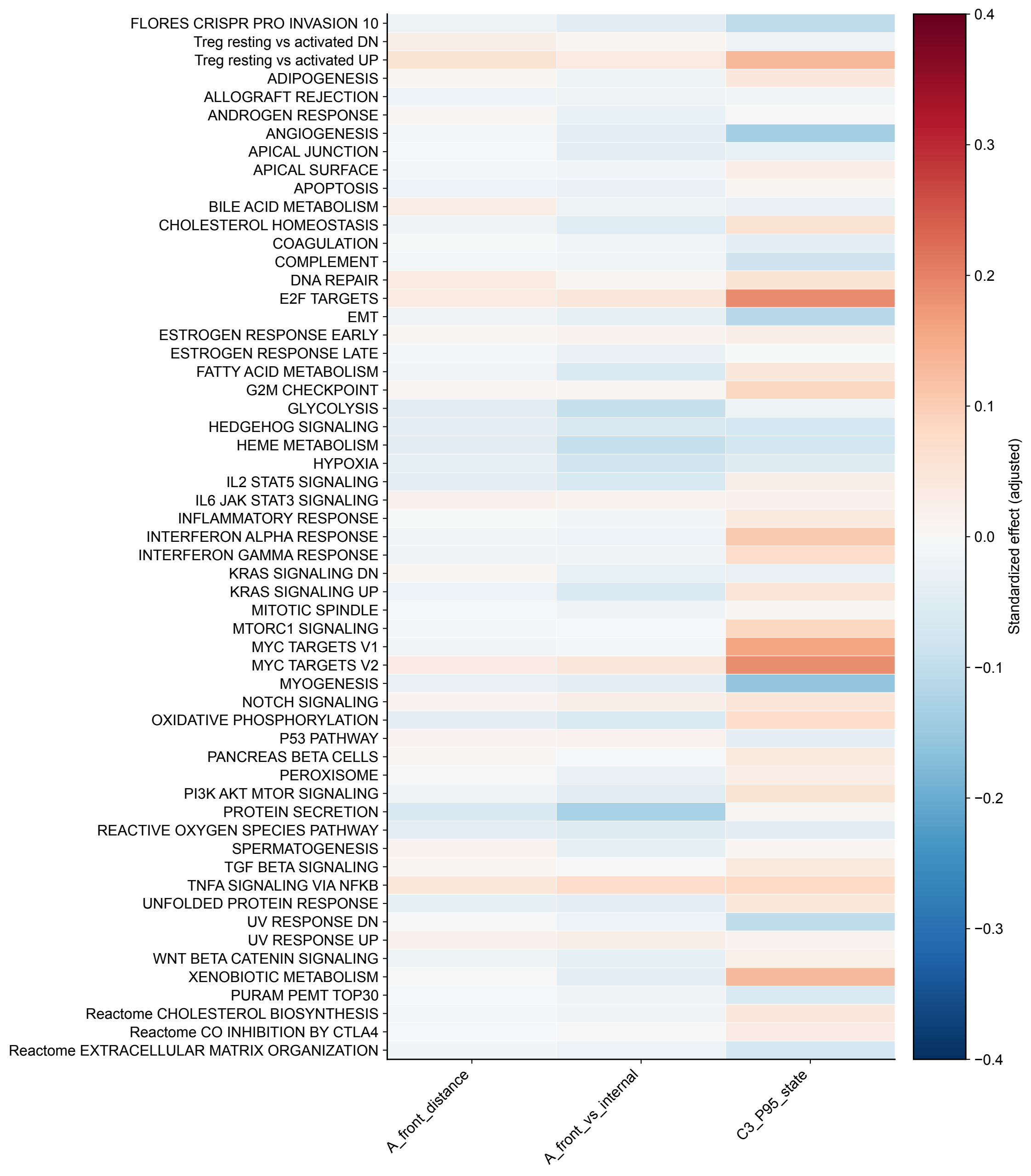

REACTIVE OXYGEN SPECIES PATHWAY
DNA REPAIR
MTORC1 SIGNALING
PI3K AKT MTOR SIGNALING
UCell / dataset-equal mean
Distance to reference set (μm)
Coverage: sections 45–51; datasets 6–6
P95 matched front / tissue
control front / tissue
P95 matched front / Epi
control front / Epi


ANDROGEN RESPONSE
UCell / dataset-equal mean
0.135
0.130
0.125
0.120
0.115
0.110
Coverage: sections 45–51; datasets 6–6
0
200
400
600
800
1000
Distance to reference set (μm)


PURAM PEMT TOP30
UCell / dataset-equal mean
0.16
0.15
0.14
0.13
0.12
0.11
Coverage: sections 45–51; datasets 6–6
0
200
400
600
800
1000
Distance to reference set (μm)


FLORES CRISPR PRO INVASION 10
UCell / dataset-equal mean
0.043
0.042
0.041
0.040
0.039
0.038
0.037
Coverage: sections 45–51; datasets 6–6
0
200
400
600
800
1000
Distance to reference set (μm)

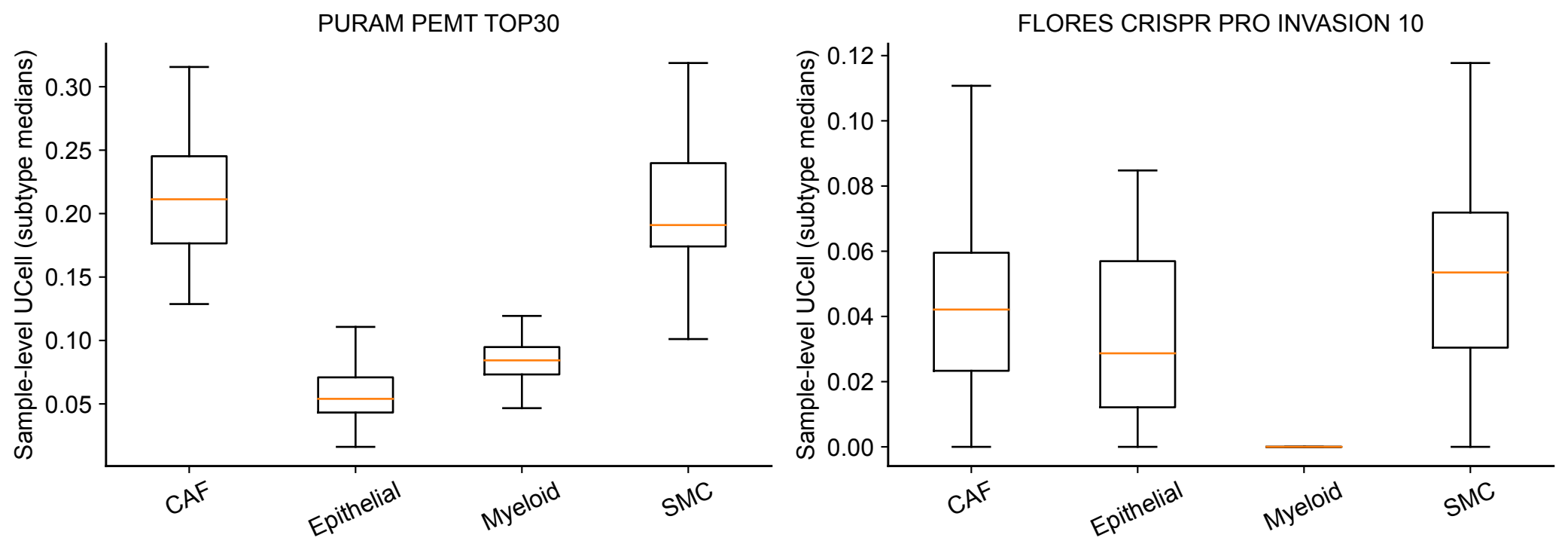

P95 matched front / tissue
control front / tissue
P95 matched front / Epi
control front / Epi


FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
APOPTOSIS
BILE ACID METABOLISM
CHOLESTEROL HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA REPAIR
E2F TARGETS
EMT
ESTROGEN RESPONSE EARLY
ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
G2M CHECKPOINT
GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
IL2 STAT5 SIGNALING
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
MTORC1 SIGNALING
MYC TARGETS V1
MYC TARGETS V2
MYOGENESIS
NOTCH SIGNALING
OXIDATIVE PHOSPHORYLATION
P53 PATHWAY
PANCREAS BETA CELLS
PEROXISOME
PI3K AKT MTOR SIGNALING
PROTEIN SECRETION
REACTIVE OXYGEN SPECIES PATHWAY
SPERMATOGENESIS
TGF BETA SIGNALING
TNFA SIGNALING VIA NFKB
UNFOLDED PROTEIN RESPONSE
UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
PURAM PEMT TOP30
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION
A_front_distance
A_front_vs_internal
C3_P95_state
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Standardized effect (adjusted)

FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
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ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
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GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
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IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
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MYC TARGETS V1
MYC TARGETS V2
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UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
PURAM PEMT TOP30
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION

A_front_vs_deep
A_front_vs_near

0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Adjusted front-minus-interior effect (SD)

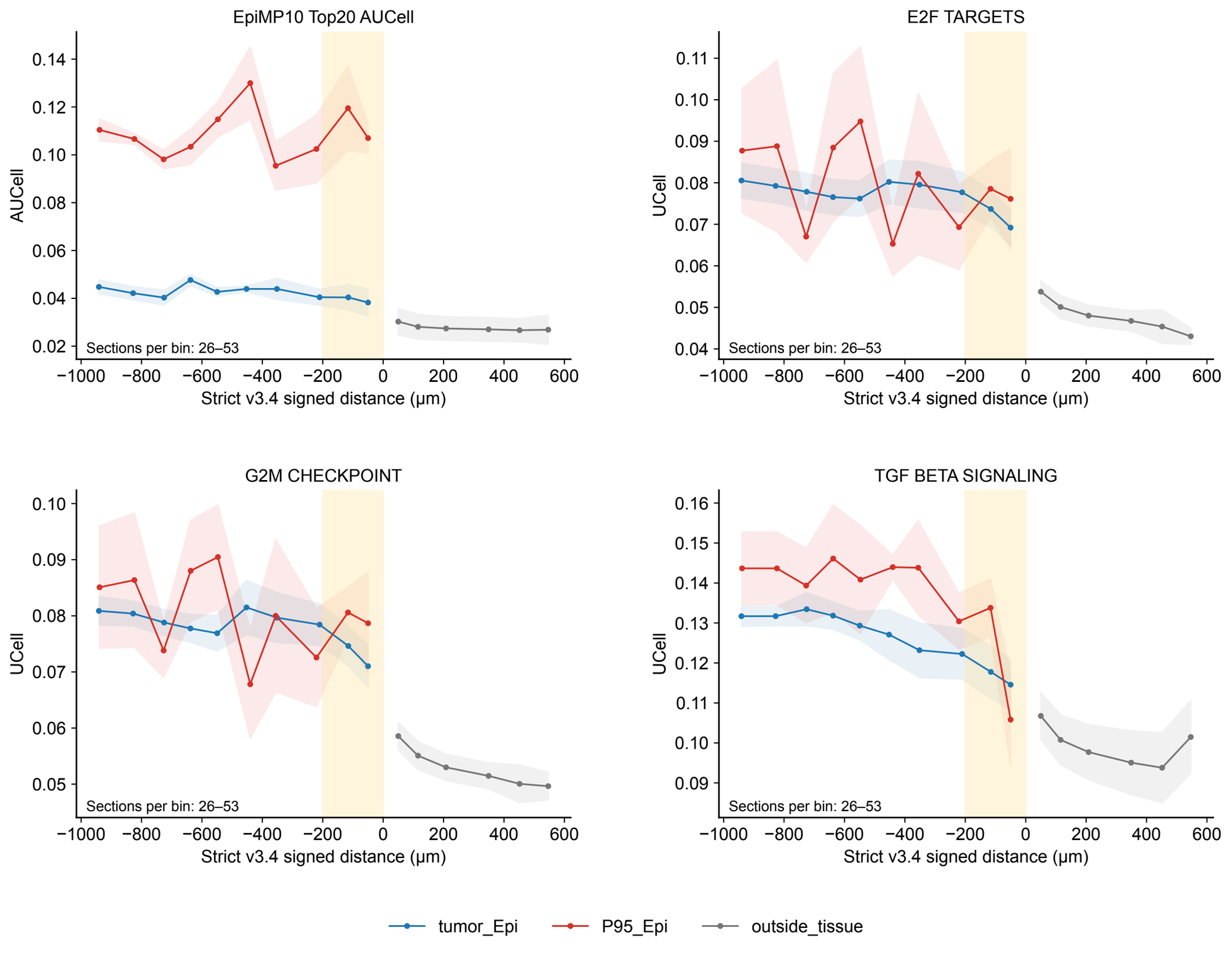

PURAM PEMT TOP30
Sample-level UCell (subtype medians)
0.30
0.25
0.20
0.15
0.10
0.05
CAF
Epithelial
Myeloid
SMC
FLORES CRISPR PRO INVASION 10
Sample-level UCell (subtype medians)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
CAF
Epithelial
Myeloid
SMC

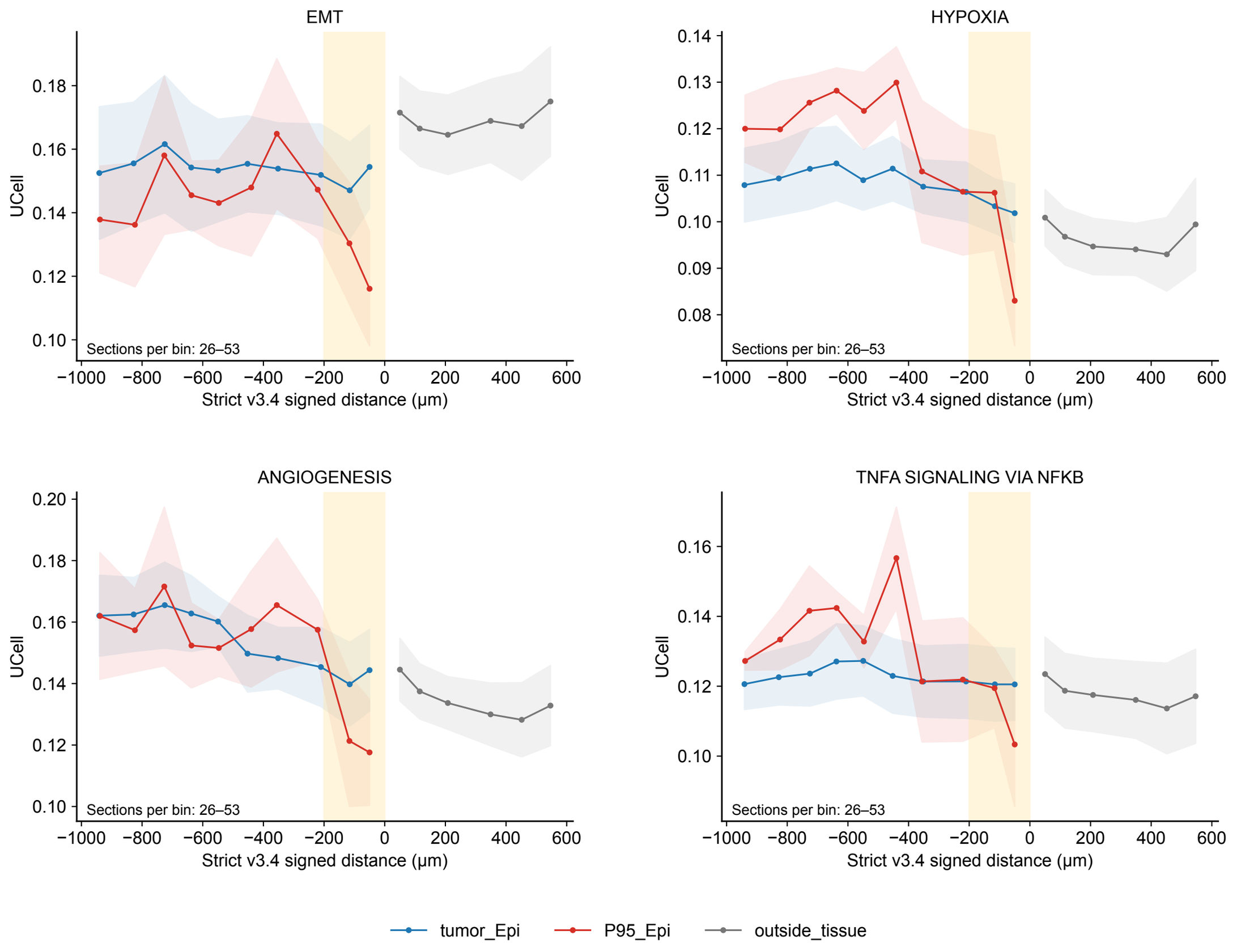
FLORES CRISPR PRO INVASION 10
Treg resting vs activated DN
Treg resting vs activated UP
ADIPOGENESIS
ALLOGRAFT REJECTION
ANDROGEN RESPONSE
ANGIOGENESIS
APICAL JUNCTION
APICAL SURFACE
APOPTOSIS
BILE ACID METABOLISM
CHOLESTEROL HOMEOSTASIS
COAGULATION
COMPLEMENT
DNA REPAIR
E2F TARGETS
EMT
ESTROGEN RESPONSE EARLY
ESTROGEN RESPONSE LATE
FATTY ACID METABOLISM
G2M CHECKPOINT
GLYCOLYSIS
HEDGEHOG SIGNALING
HEME METABOLISM
HYPOXIA
IL2 STAT5 SIGNALING
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
KRAS SIGNALING DN
KRAS SIGNALING UP
MITOTIC SPINDLE
MTORC1 SIGNALING
MYC TARGETS V1
MYC TARGETS V2
MYOGENESIS
NOTCH SIGNALING
OXIDATIVE PHOSPHORYLATION
P53 PATHWAY
PANCREAS BETA CELLS
PEROXISOME
PI3K AKT MTOR SIGNALING
PROTEIN SECRETION
REACTIVE OXYGEN SPECIES PATHWAY
SPERMATOGENESIS
TGF BETA SIGNALING
TNFA SIGNALING VIA NFKB
UNFOLDED PROTEIN RESPONSE
UV RESPONSE DN
UV RESPONSE UP
WNT BETA CATENIN SIGNALING
XENOBIOTIC METABOLISM
Reactome CHOLESTEROL BIOSYNTHESIS
Reactome CO INHIBITION BY CTLA4
Reactome EXTRACELLULAR MATRIX ORGANIZATION

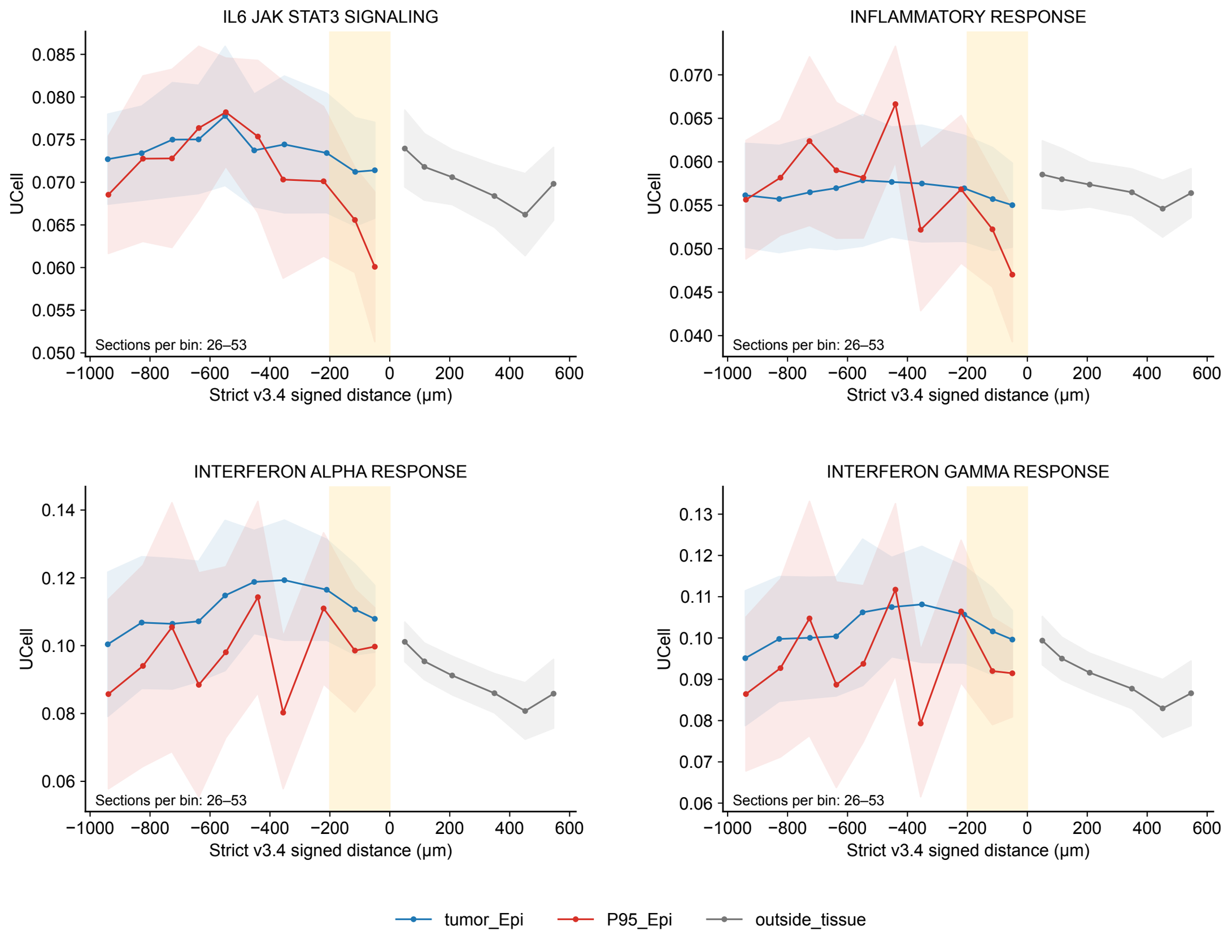
effect

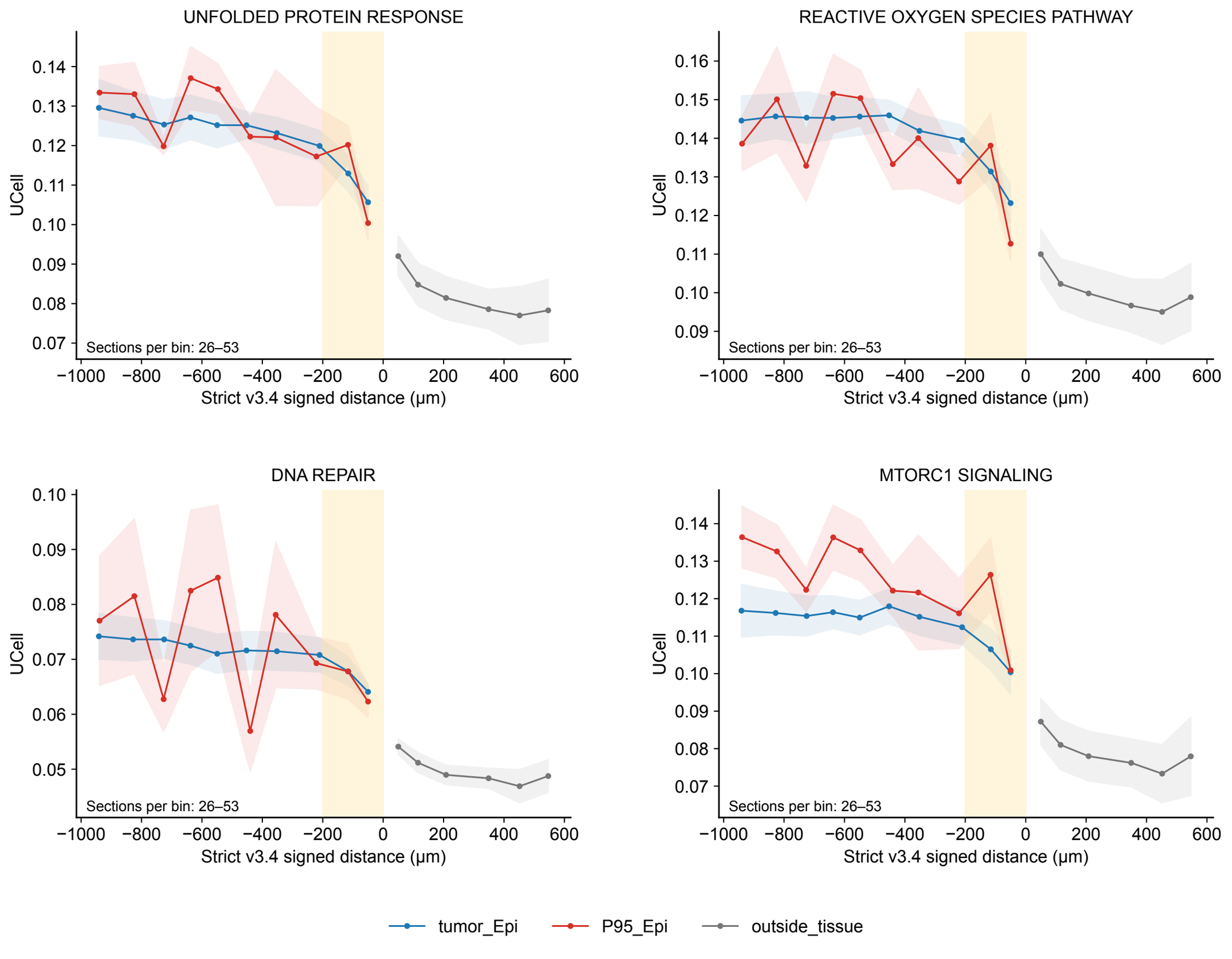
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

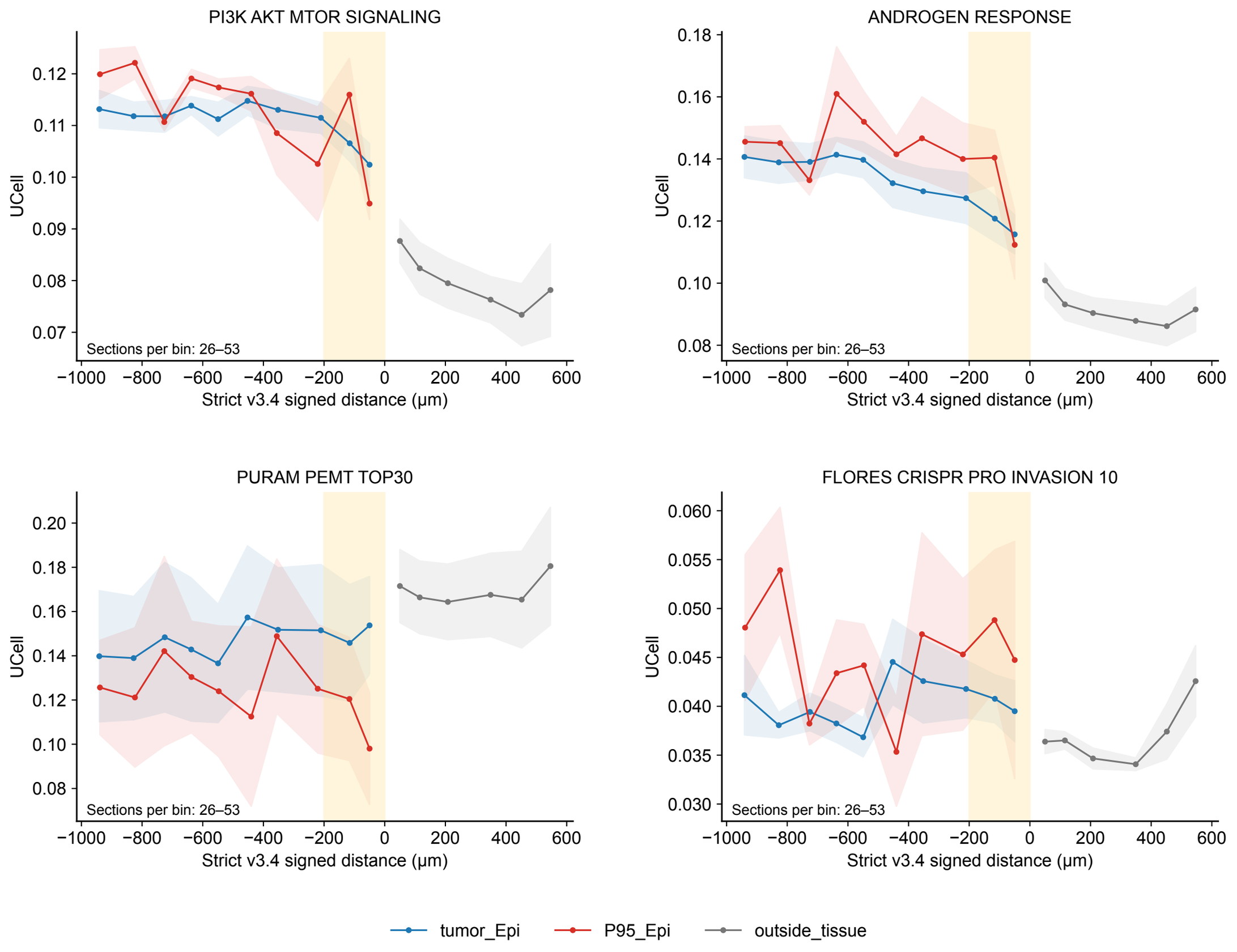
pEMT-high versus other front (adjusted SD)

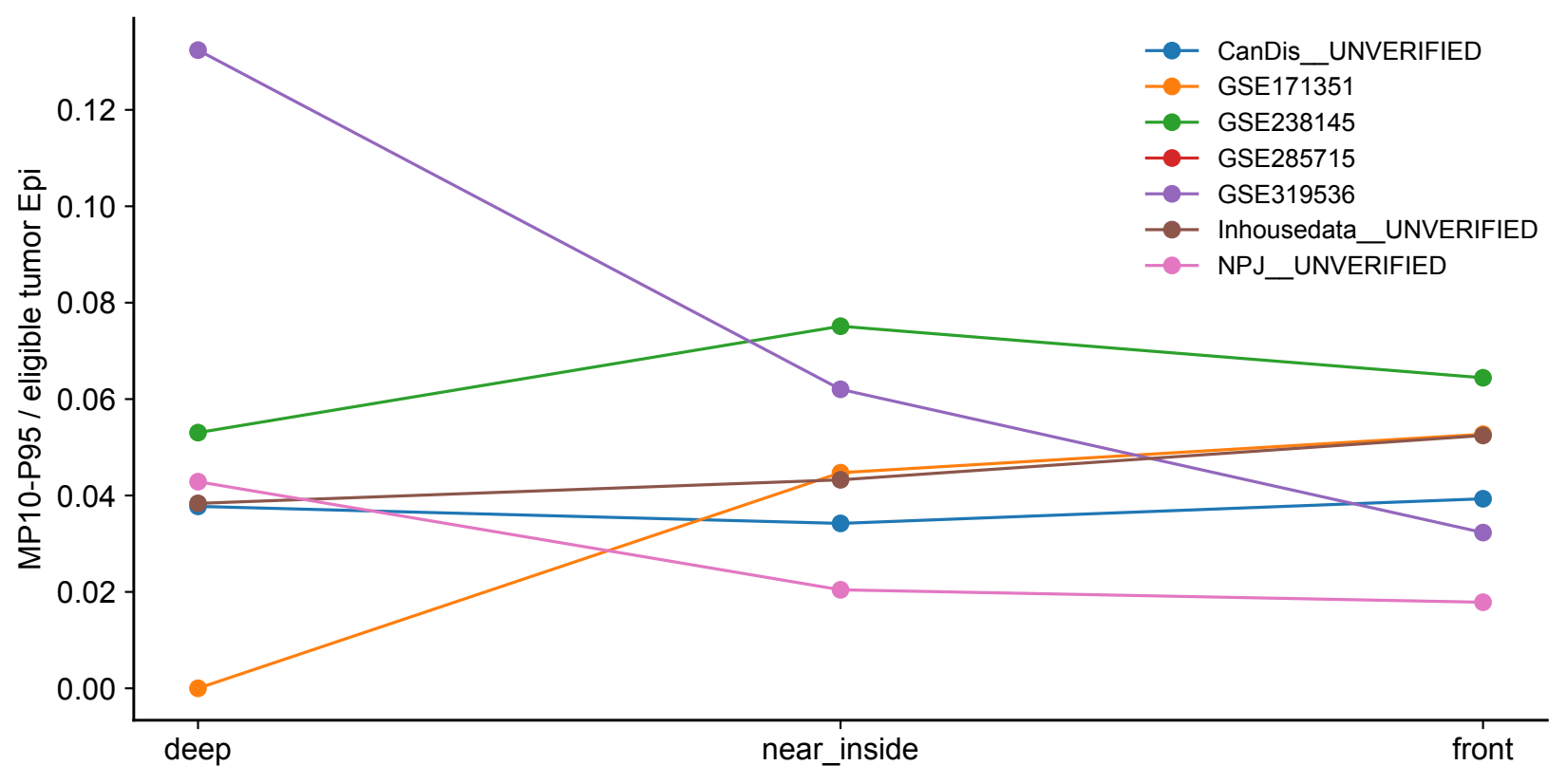

EpiMP10 Top20 AUCell
AUCell
Sections per bin: 26–53
Strict v3.4 signed distance (μm)
E2F TARGETS
UCell
Sections per bin: 26–53
Strict v3.4 signed distance (μm)
G2M CHECKPOINT
UCell
Sections per bin: 26–53
Strict v3.4 signed distance (μm)
TGF BETA SIGNALING
UCell
Sections per bin: 26–53
Strict v3.4 signed distance (μm)
tumor_Epi
P95_Epi
outside_tissue

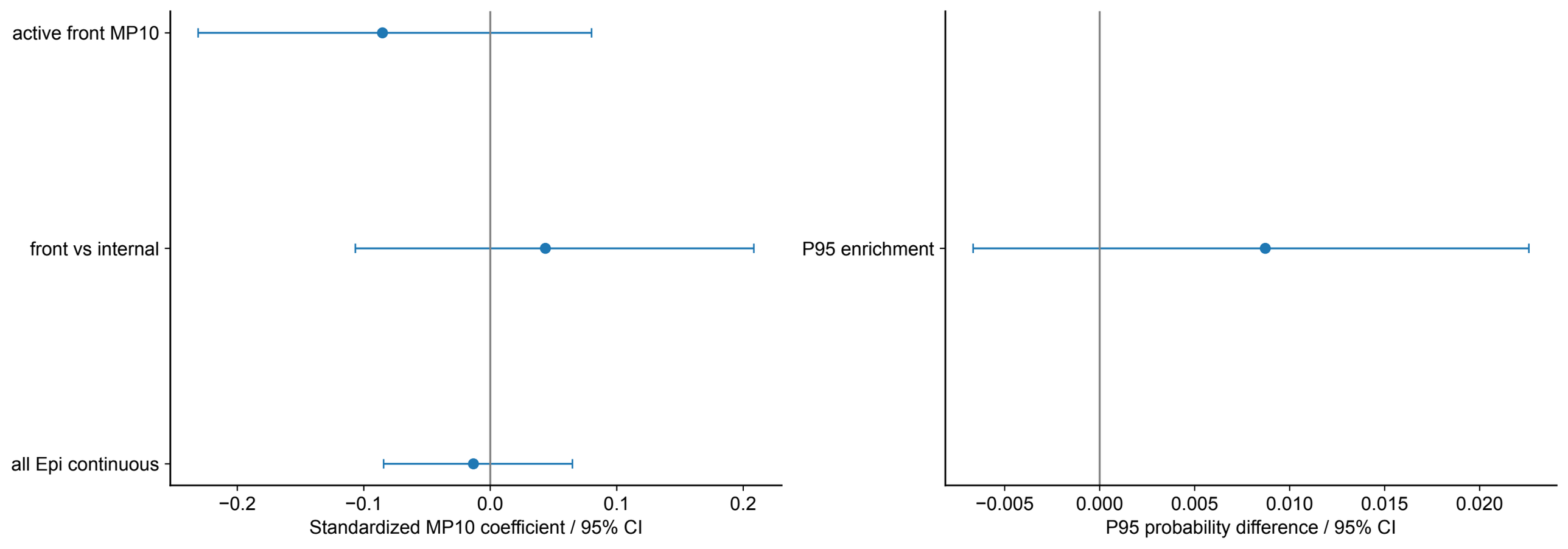

EMT
HYPOXIA
ANGIOGENESIS
TNFA SIGNALING VIA NFKB
UCell
Strict v3.4 signed distance (μm)
Sections per bin: 26–53
tumor_Epi
P95_Epi
outside_tissue

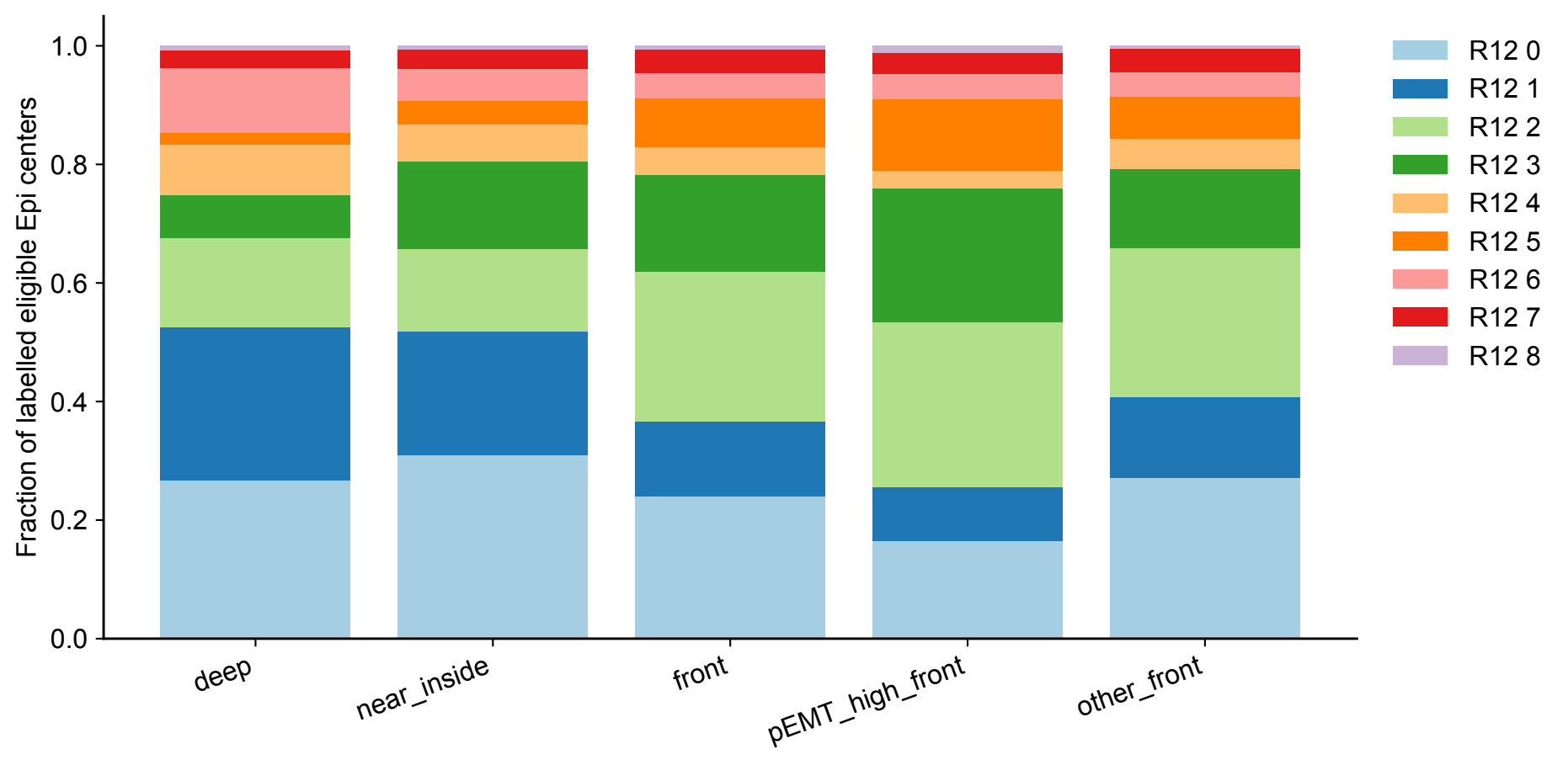

IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
UCell
Strict v3.4 signed distance (μm)
Sections per bin: 26–53
tumor_Epi
P95_Epi
outside_tissue

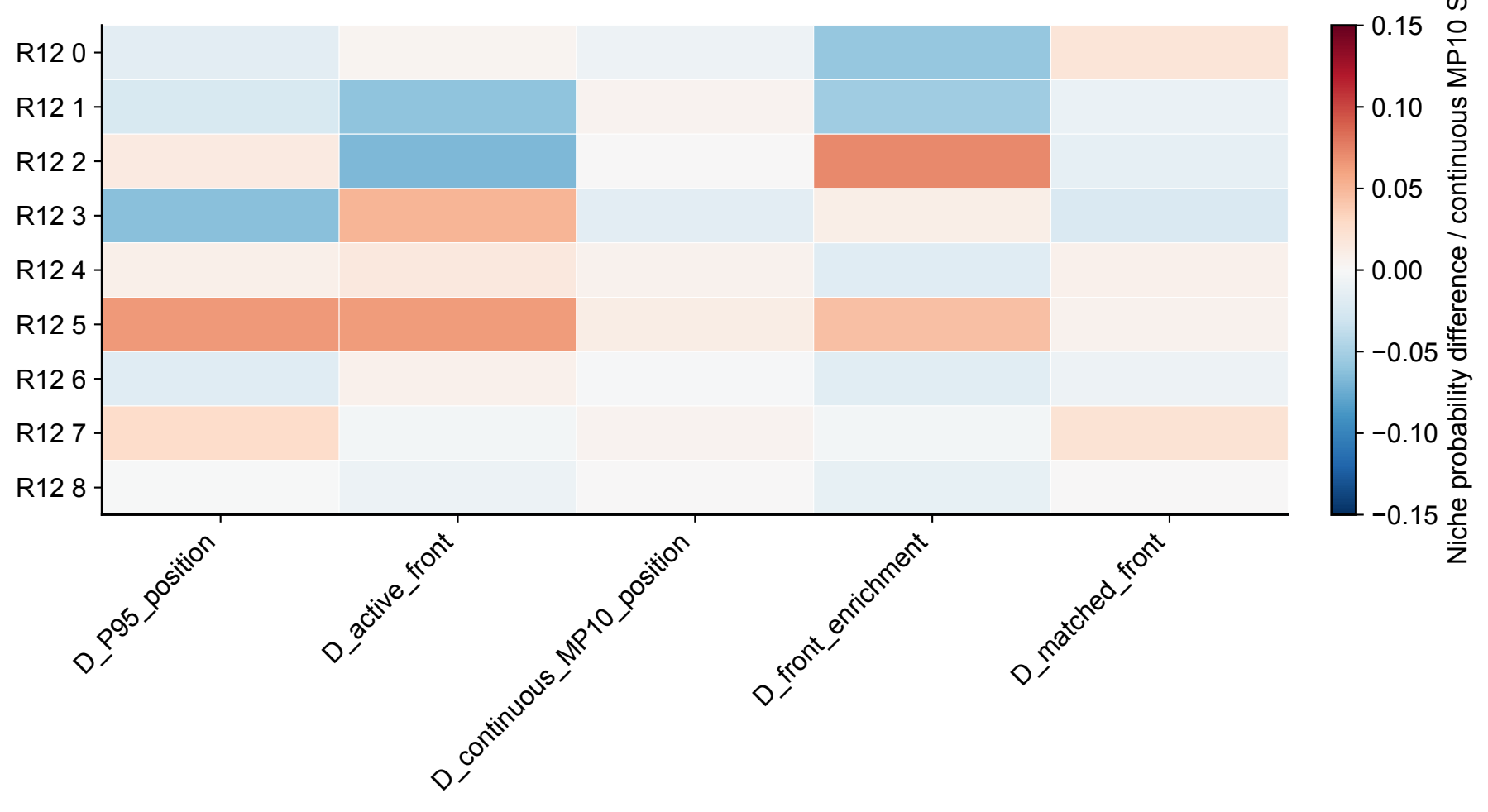

UNFOLDED PROTEIN RESPONSE
REACTIVE OXYGEN SPECIES PATHWAY
DNA REPAIR
MTORC1 SIGNALING
UCell
Strict v3.4 signed distance (μm)
Sections per bin: 26–53
tumor_Epi
P95_Epi
outside_tissue

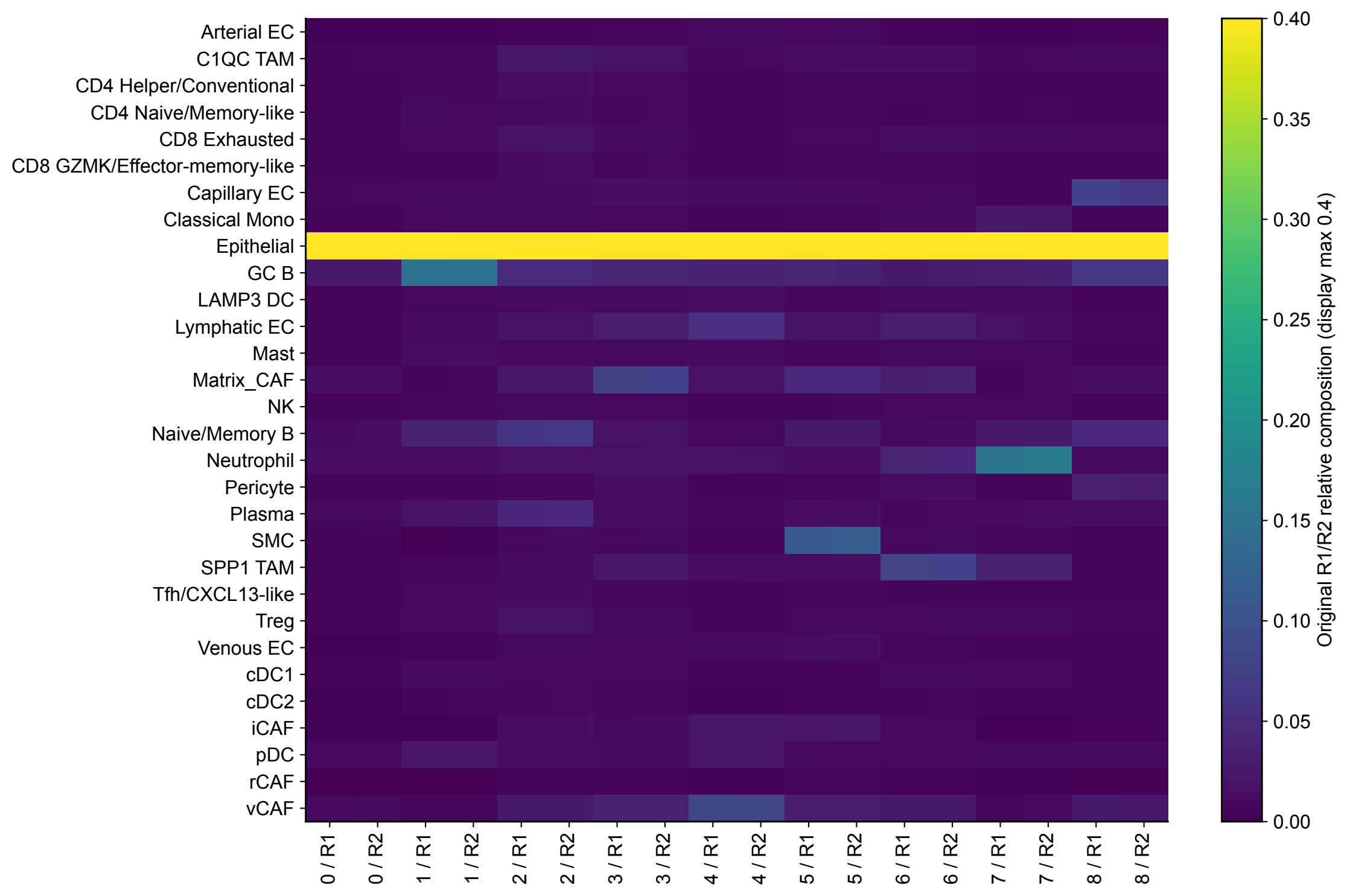

PI3K AKT MTOR SIGNALING
ANDROGEN RESPONSE
PURAM PEMT TOP30
FLORES CRISPR PRO INVASION 10
UCell
Strict v3.4 signed distance (μm)
Sections per bin: 26–53
tumor_Epi
P95_Epi
outside_tissue


MP10-P95 / eligible tumor Epi
0.12
0.10
0.08
0.06
0.04
0.02
0.00
deep
near_inside
front
CanDis__UNVERIFIED
GSE171351
GSE238145
GSE285715
GSE319536
Inhousedata__UNVERIFIED
NPJ__UNVERIFIED


active front MP10
front vs internal
all Epi continuous
−0.2
−0.1
0.0
0.1
0.2
Standardized MP10 coefficient / 95% CI
P95 enrichment
−0.005
0.000
0.005
0.010
0.015
0.020
P95 probability difference / 95% CI

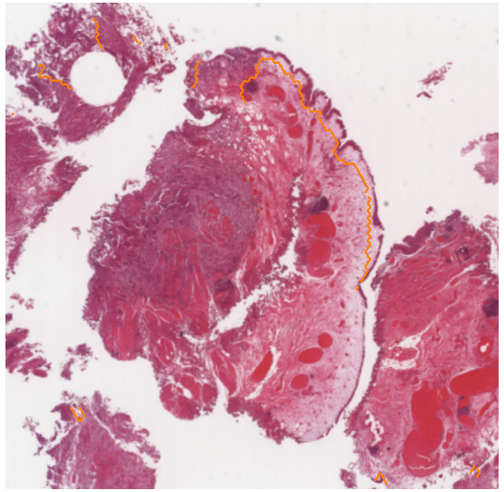

Fraction of labelled eligible Epi centers
1.0
0.8
0.6
0.4
0.2
0.0
deep
near_inside
front
pEMT_high_front
other_front
R12 0
R12 1
R12 2
R12 3
R12 4
R12 5
R12 6
R12 7
R12 8


R12 0
R12 1
R12 2
R12 3
R12 4
R12 5
R12 6
R12 7
R12 8
D_P95_position
D_active_front
D_continuous_MP10_position
D_front_enrichment
D_matched_front
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Niche probability difference / continuous MP10 S

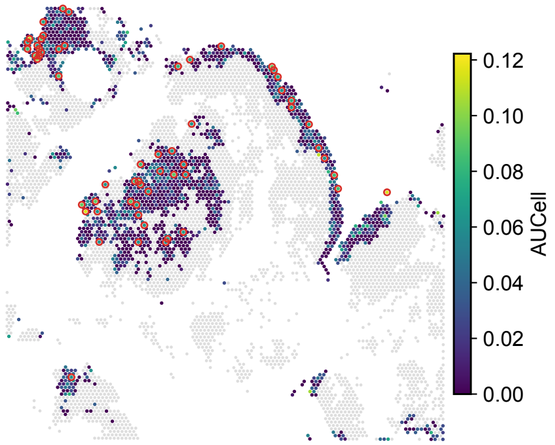

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
Epithelial
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix_CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF
0 / R1
0 / R2
1 / R1
1 / R2
2 / R1
2 / R2
3 / R1
3 / R2
4 / R1
4 / R2
5 / R1
5 / R2
6 / R1
6 / R2
7 / R1
7 / R2
8 / R1
8 / R2
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Original R1/R2 relative composition (display max 0.4)

GSE319536_BC01 | frozen v3.4 / Epi MP10 and literature programs

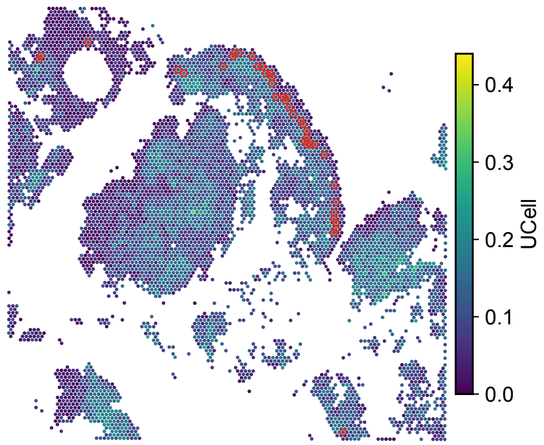
H&E / observed outer boundary



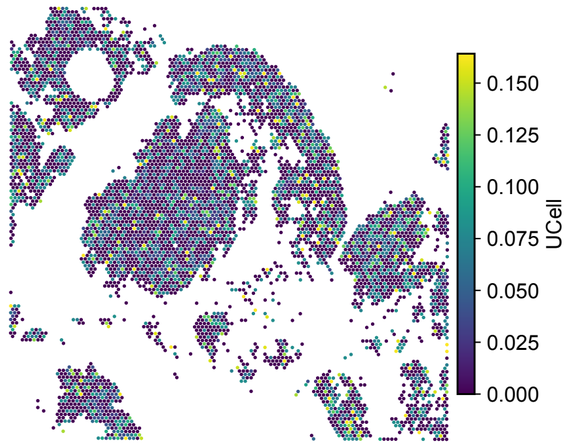
Mixed-spot MP10 AUCell / P95



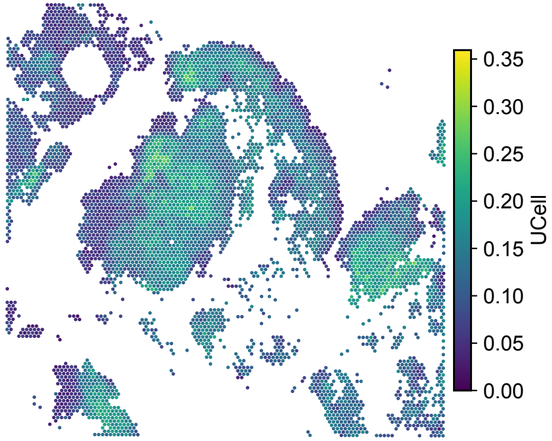
Puram p-EMT Top30 / active front



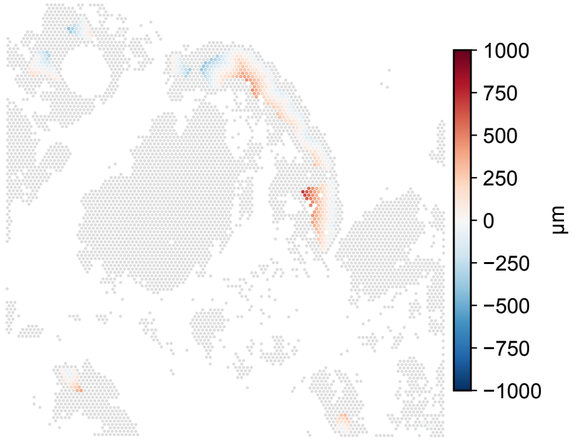
Flores positive 10



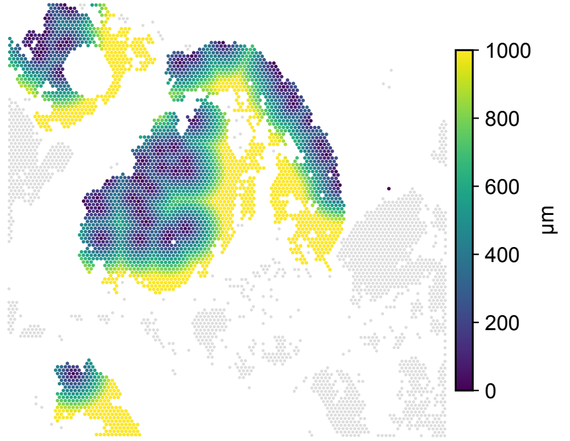
Hallmark EMT (mixed spot)



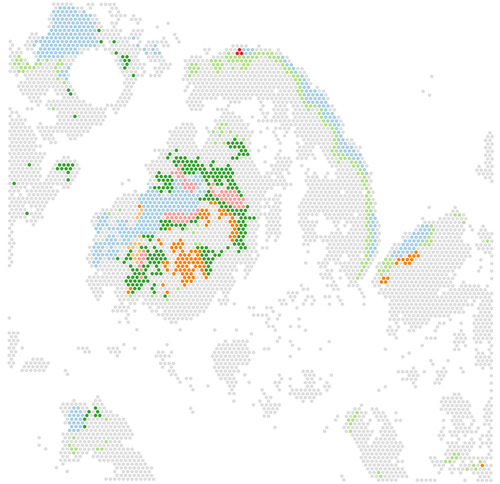
Strict v3.4 signed distance



Nearest MP10-P95 distance



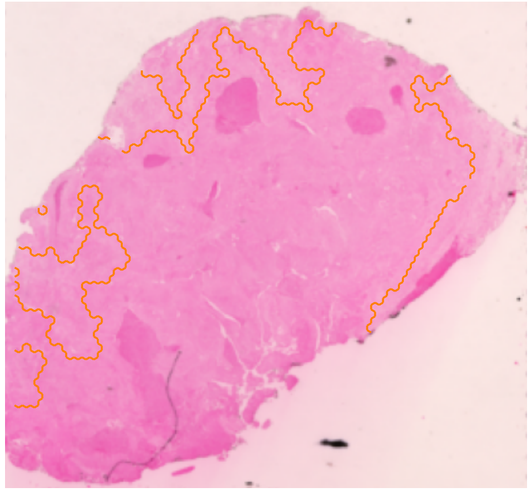
SAMPLE R12 LEIDEN 0–8



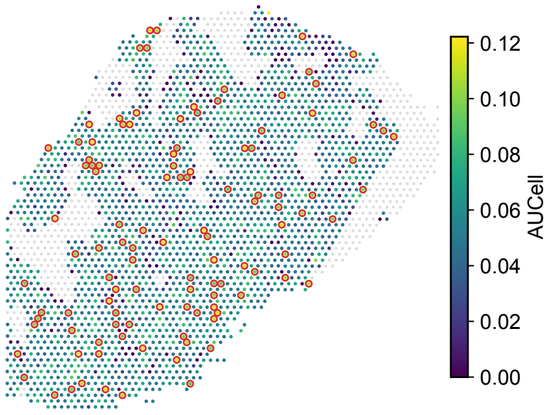
R12 0 R12 1 R12 2 R12 3 R12 4 R12 5 R12 6 R12 7 R12 8 P95 / pEMT-high front

CanDis_2026_Sample9 | frozen v3.4 / Epi MP10 and literature programs

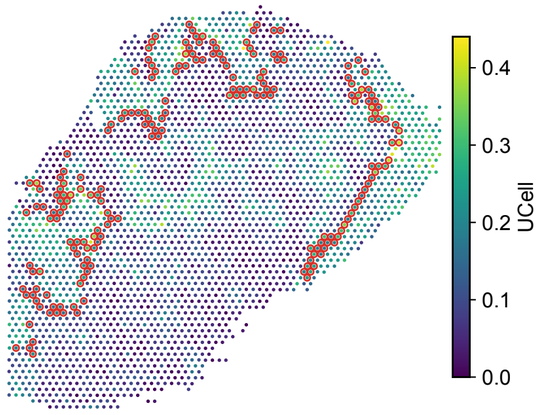
H&E / observed outer boundary



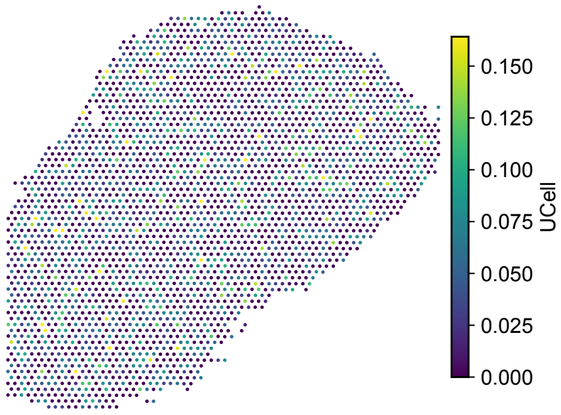
Mixed-spot MP10 AUCell / P95



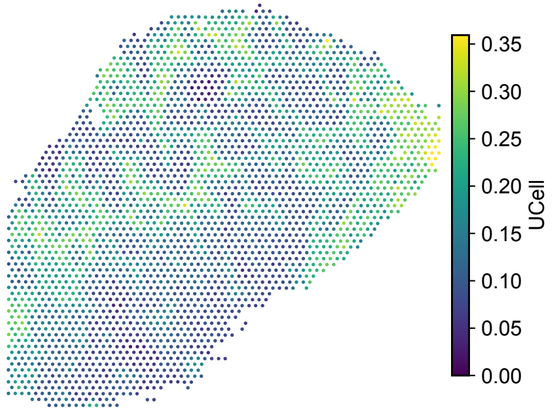
Puram p-EMT Top30 / active front



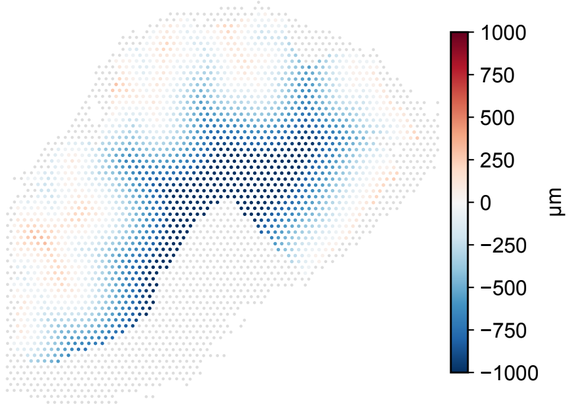
Flores positive 10



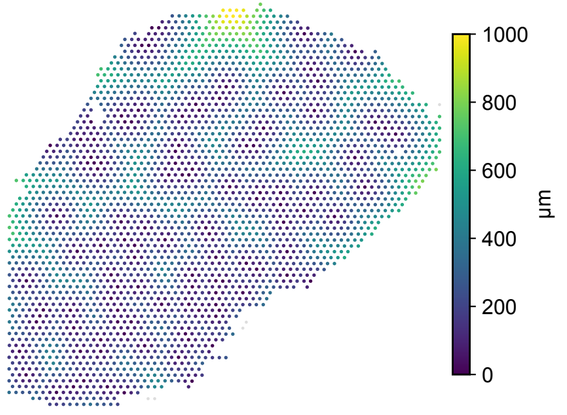
Hallmark EMT (mixed spot)



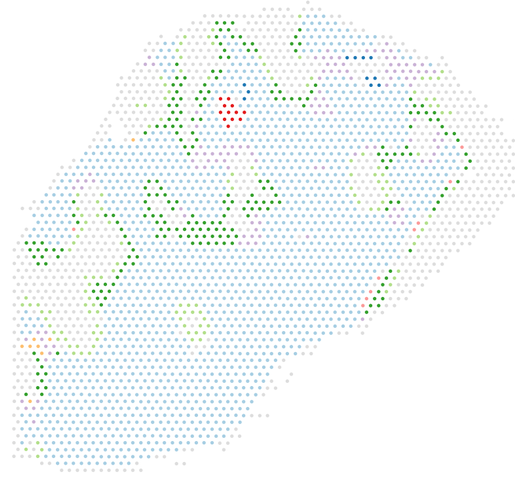
Strict v3.4 signed distance



Nearest MP10-P95 distance



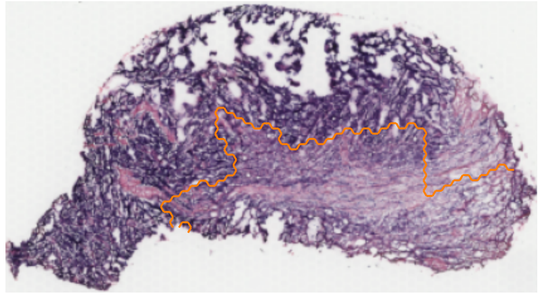
SAMPLE R12 LEIDEN 0–8



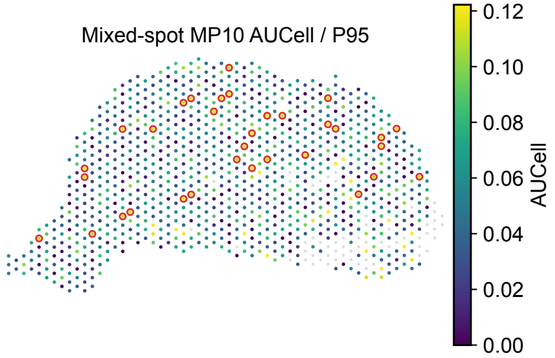
R12 0 R12 1 R12 2 R12 3 R12 4 R12 5 R12 6 R12 7 R12 8 P95 / pEMT-high front

GSE171351_B1 | frozen v3.4 / Epi MP10 and literature programs

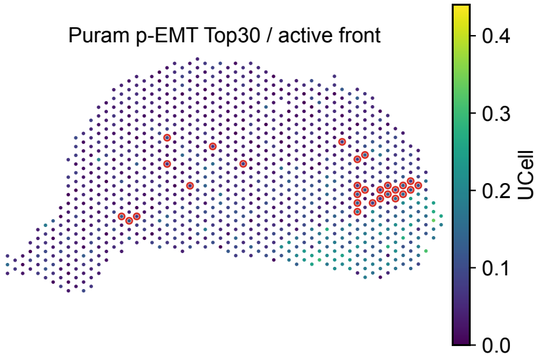
H&E / observed outer boundary



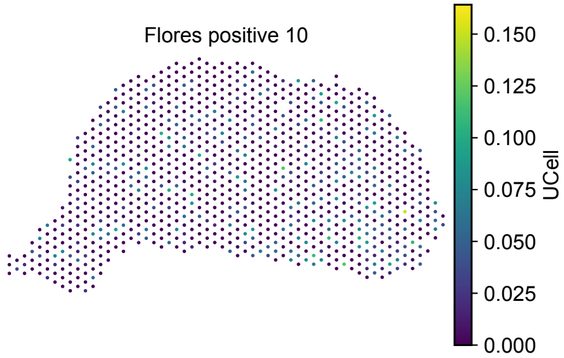
Mixed-spot MP10 AUCell / P95



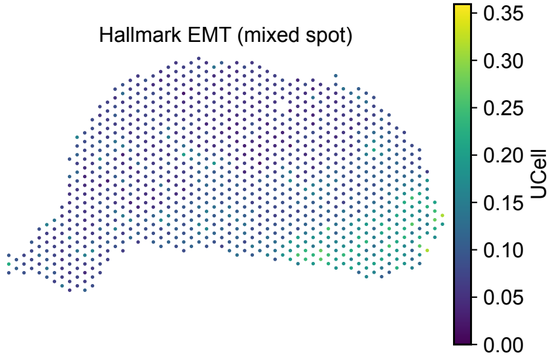
Puram p-EMT Top30 / active front



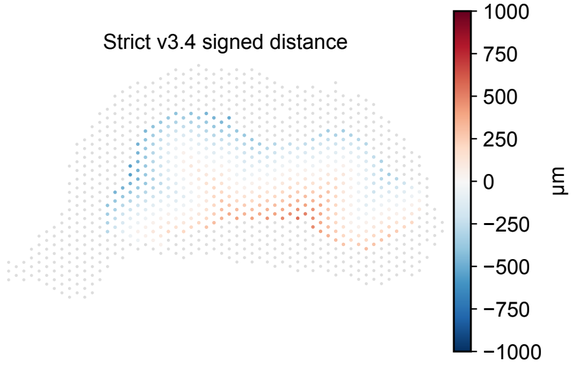
Flores positive 10



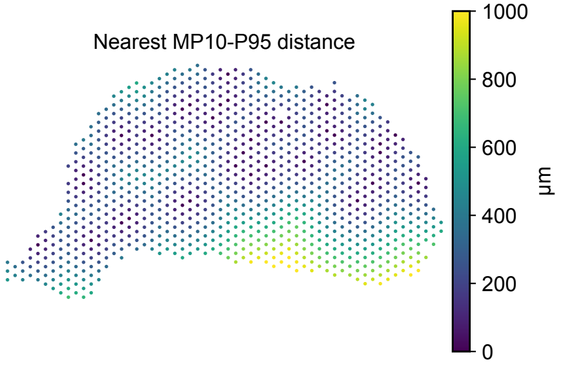
Hallmark EMT (mixed spot)



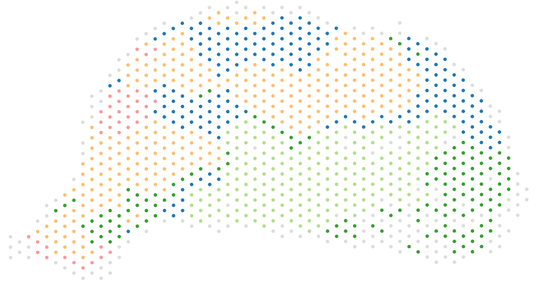
Strict v3.4 signed distance



Nearest MP10-P95 distance



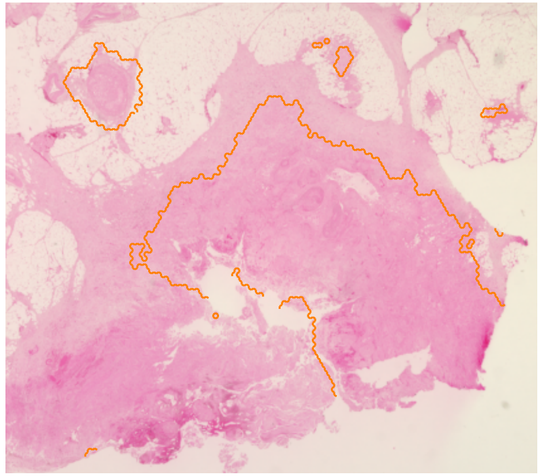
SAMPLE R12 LEIDEN 0–8



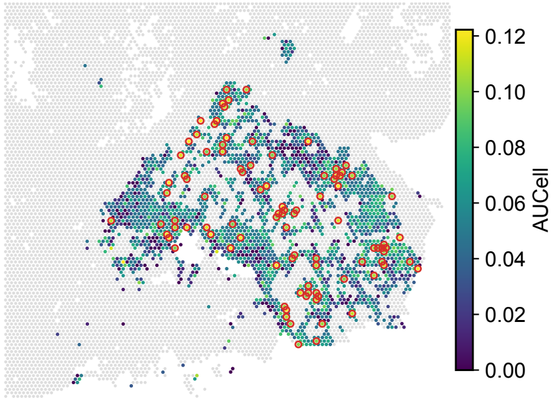
R12 0 R12 1 R12 2 R12 3 R12 4 R12 5 R12 6 R12 7 R12 8 P95 / pEMT-high front

GSE238145_B14 | frozen v3.4 / Epi MP10 and literature programs

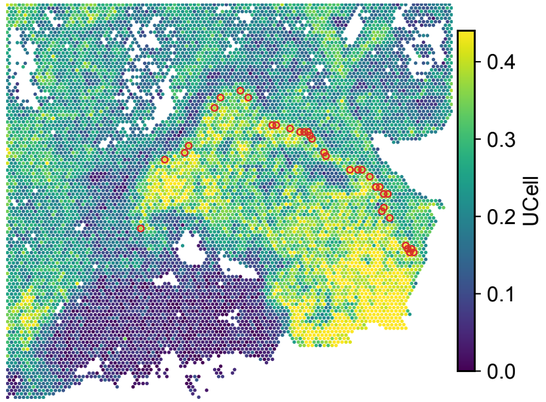
H&E / observed outer boundary



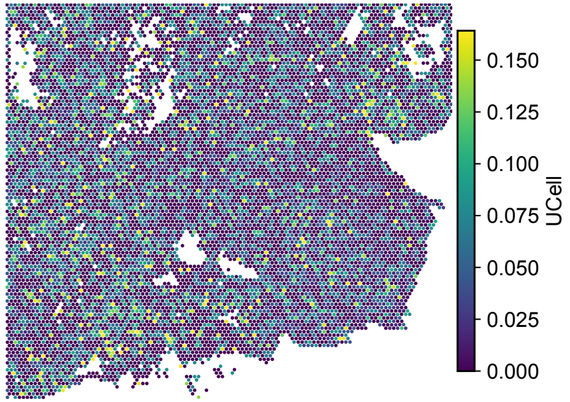
Mixed-spot MP10 AUCell / P95



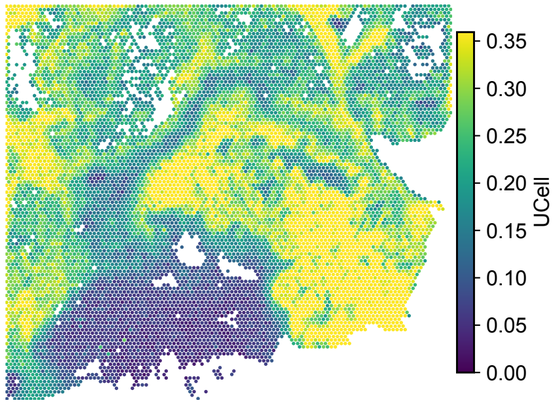
Puram p-EMT Top30 / active front



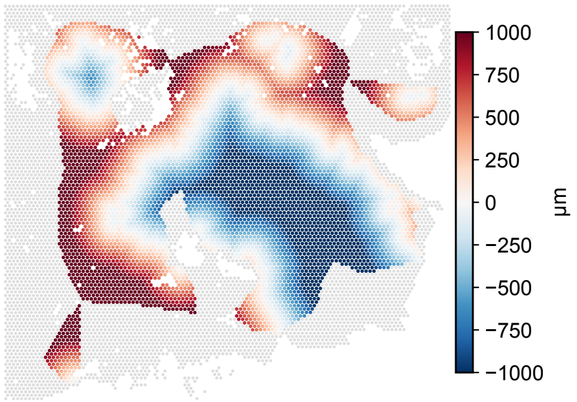
Flores positive 10



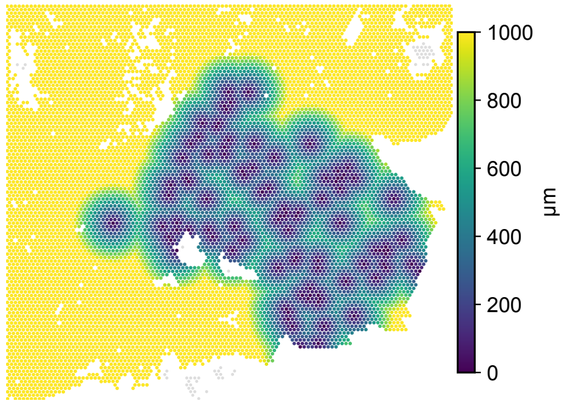
Hallmark EMT (mixed spot)



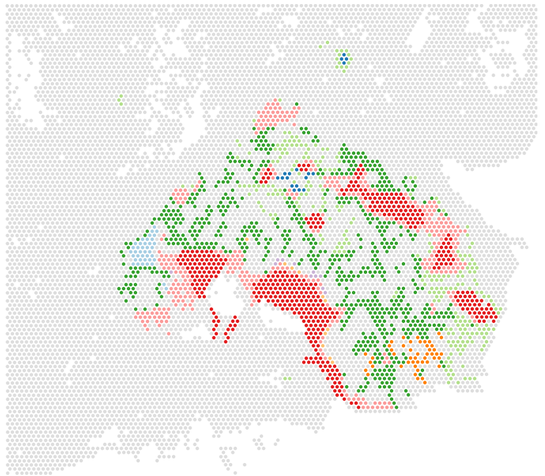
Strict v3.4 signed distance



Nearest MP10-P95 distance

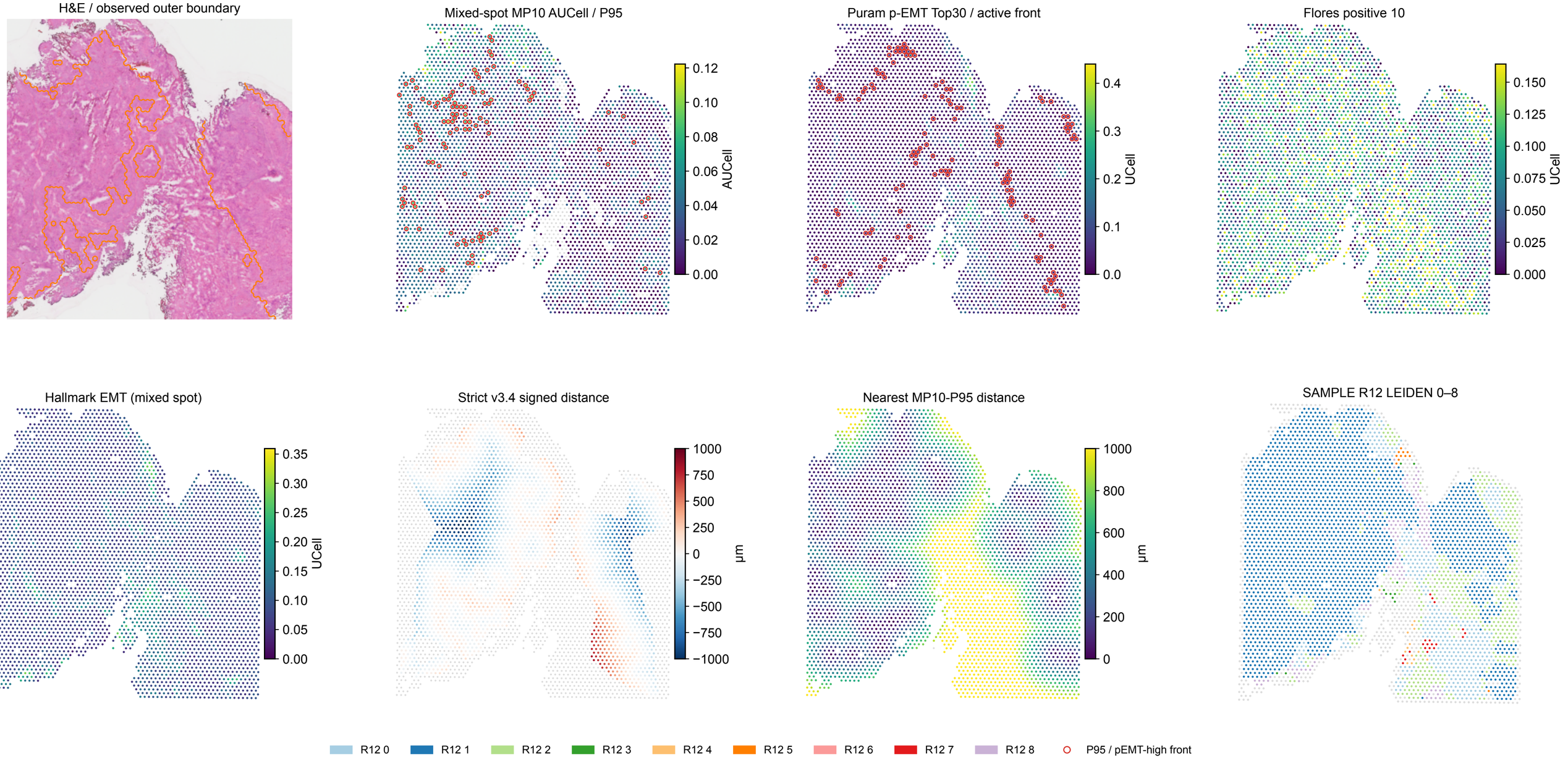


SAMPLE R12 LEIDEN 0–8



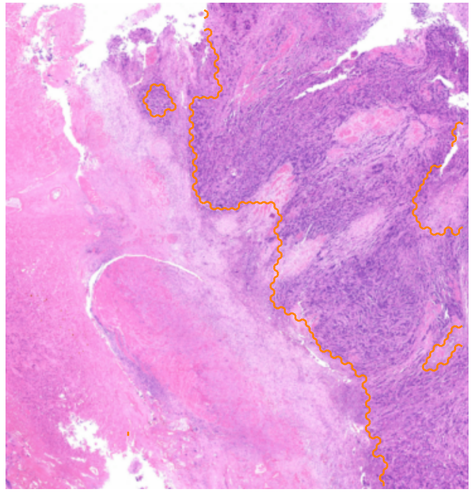
R12 0 R12 1 R12 2 R12 3 R12 4 R12 5 R12 6 R12 7 R12 8 P95 / pEMT-high front

GSE285715_T2 | frozen v3.4 / Epi MP10 and literature programs

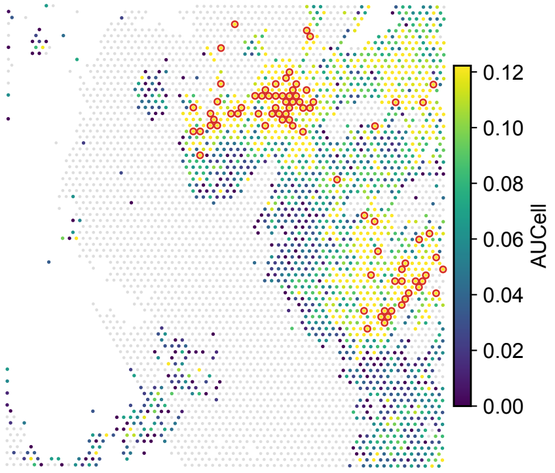


Inhousedata_T4469479 | frozen v3.4 / Epi MP10 and literature programs

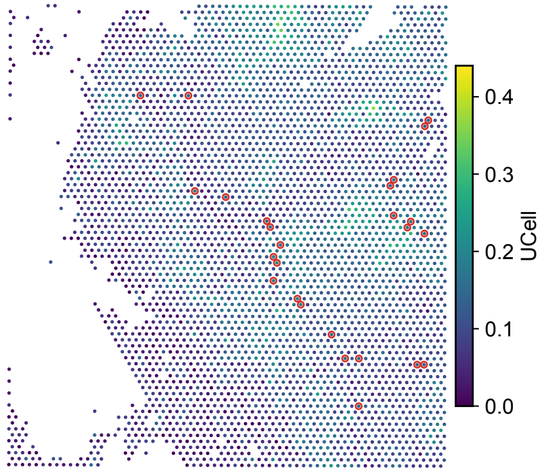
H&E / observed outer boundary



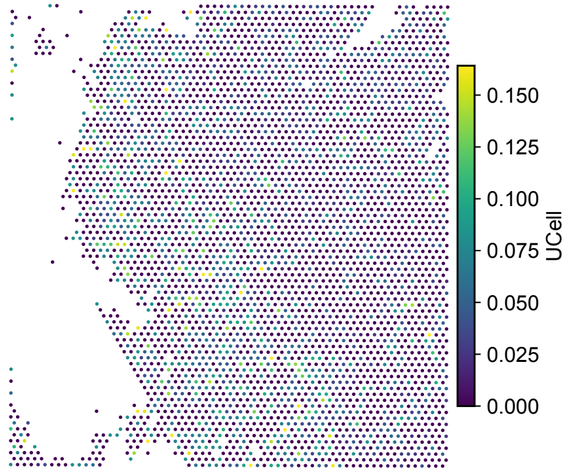
Mixed-spot MP10 AUCell / P95



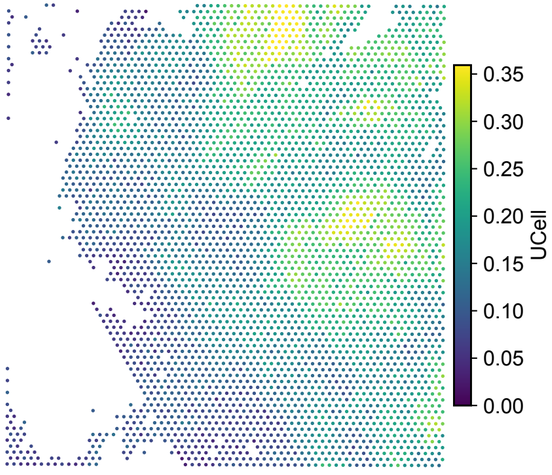
Puram p-EMT Top30 / active front



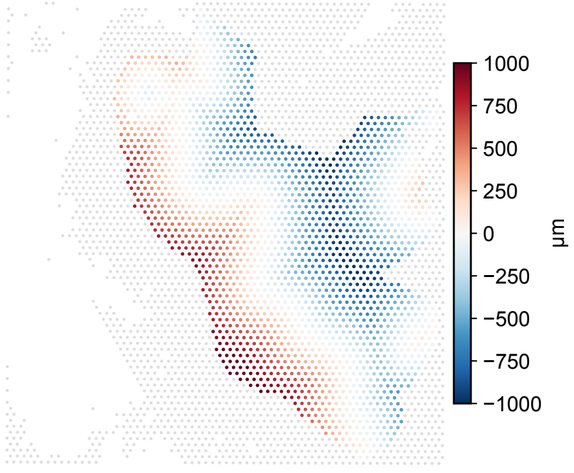
Flores positive 10



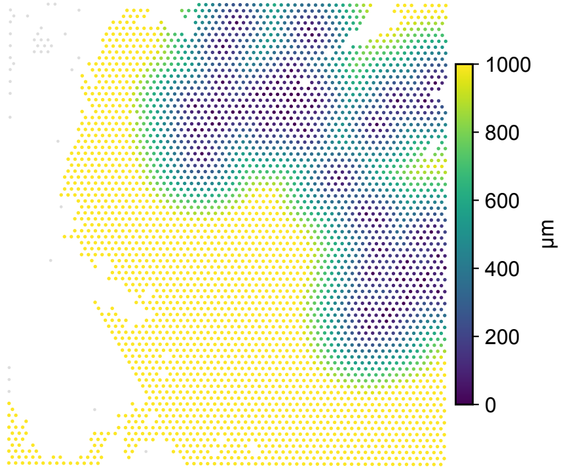
Hallmark EMT (mixed spot)



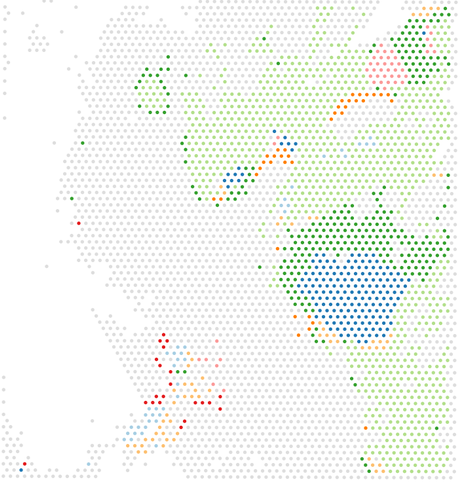
Strict v3.4 signed distance



Nearest MP10-P95 distance



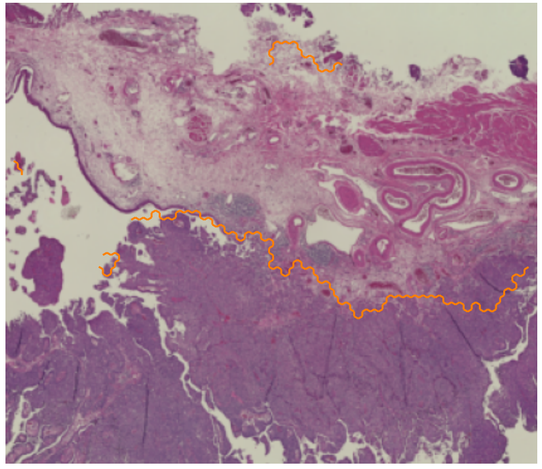
SAMPLE R12 LEIDEN 0–8



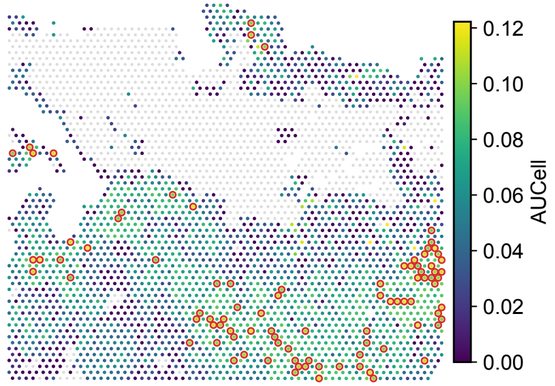
R12 0 R12 1 R12 2 R12 3 R12 4 R12 5 R12 6 R12 7 R12 8 P95 / pEMT-high front

NPJ_sample6 | frozen v3.4 / Epi MP10 and literature programs

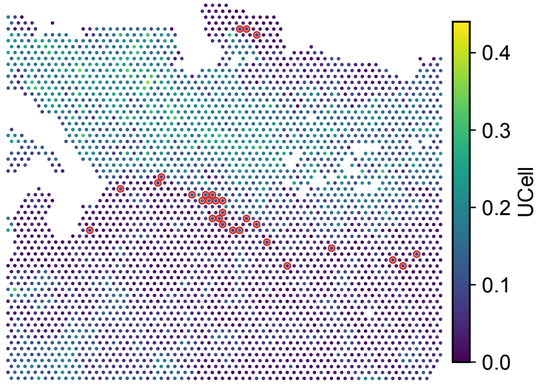
H&E / observed outer boundary



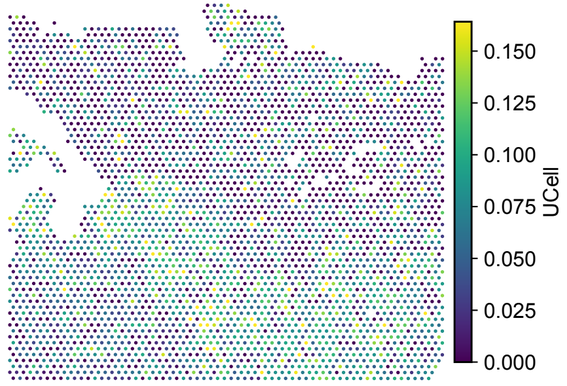
Mixed-spot MP10 AUCell / P95



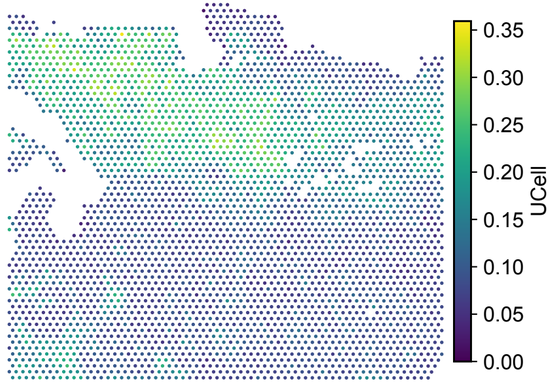
Puram p-EMT Top30 / active front



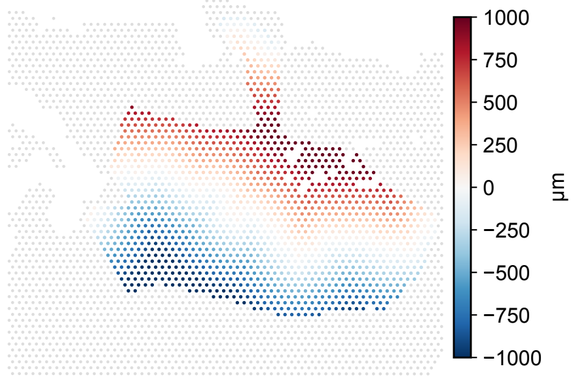
Flores positive 10



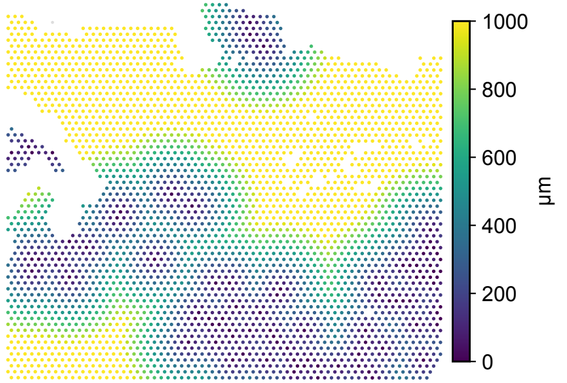
Hallmark EMT (mixed spot)



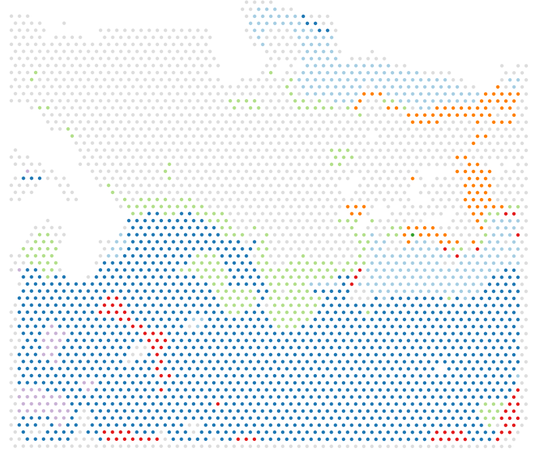
Strict v3.4 signed distance



Nearest MP10-P95 distance



SAMPLE R12 LEIDEN 0–8



R12 0 R12 1 R12 2 R12 3 R12 4 R12 5 R12 6 R12 7 R12 8 P95 / pEMT-high front